

## ENGINEERING MATHEMATICS - I

### UNIT - 4 : Multiple Integration

## ONE SHOT REVISION

### CRASH COURSE

- FULL SYLLABUS COVERAGE
- ONE SHOT VIDEO LECTURE (NEW)
- PDF NOTES (NEW)
- DPP SOLUTION (NEW)
- TOPIC WISE 10 YEAR QUESTIONS WITH SOLUTION (NEW)



By Gulshan sir



**Gateway Classes: Application  
Crash Course (New Content)  
Engineering Mathematics-I**

**Gateway Classes: You Tube  
One Shot Revision (Old)  
Engineering Mathematics-I**

## Crash Course Includes

1. FULL SYLLABUS COVERAGE
2. ONE SHOT VIDEO LECTURE (NEW)
3. PDF NOTES (NEW)
4. DPP SOLUTION (NEW)
5. TOPIC WISE 10 YEARS QUESTION WITH SOLUTION (NEW)
6. RESULT ORIENTED CONTENT

**All Subjects in Crash Course**

**Crash Course Link in Description Helpline-7819 0058 53**

## UNIT- 4 : Multiple integration

- Double integral
- Change of order of integration (7)
- Triple integral
- Change of variables
- Beta and Gama function and their properties (7)
- Dirichlet's integral and its applications to area and volume (7)
- Liouville's extensions of Dirichlet's integral





## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 1

#### Today's target

- DOUBLE INTEGRAL
- PYQs
- DPP



By Gulshan sir



➤ The integral  $\iint_R f(x, y) dA$  is called Double Integral of  $f(x, y)$  over the region  $R$ .

### Evaluation of Double Integral

**CASE – 1 :** If the region  $R$  is defined as

(i)  $a \leq x \leq b$

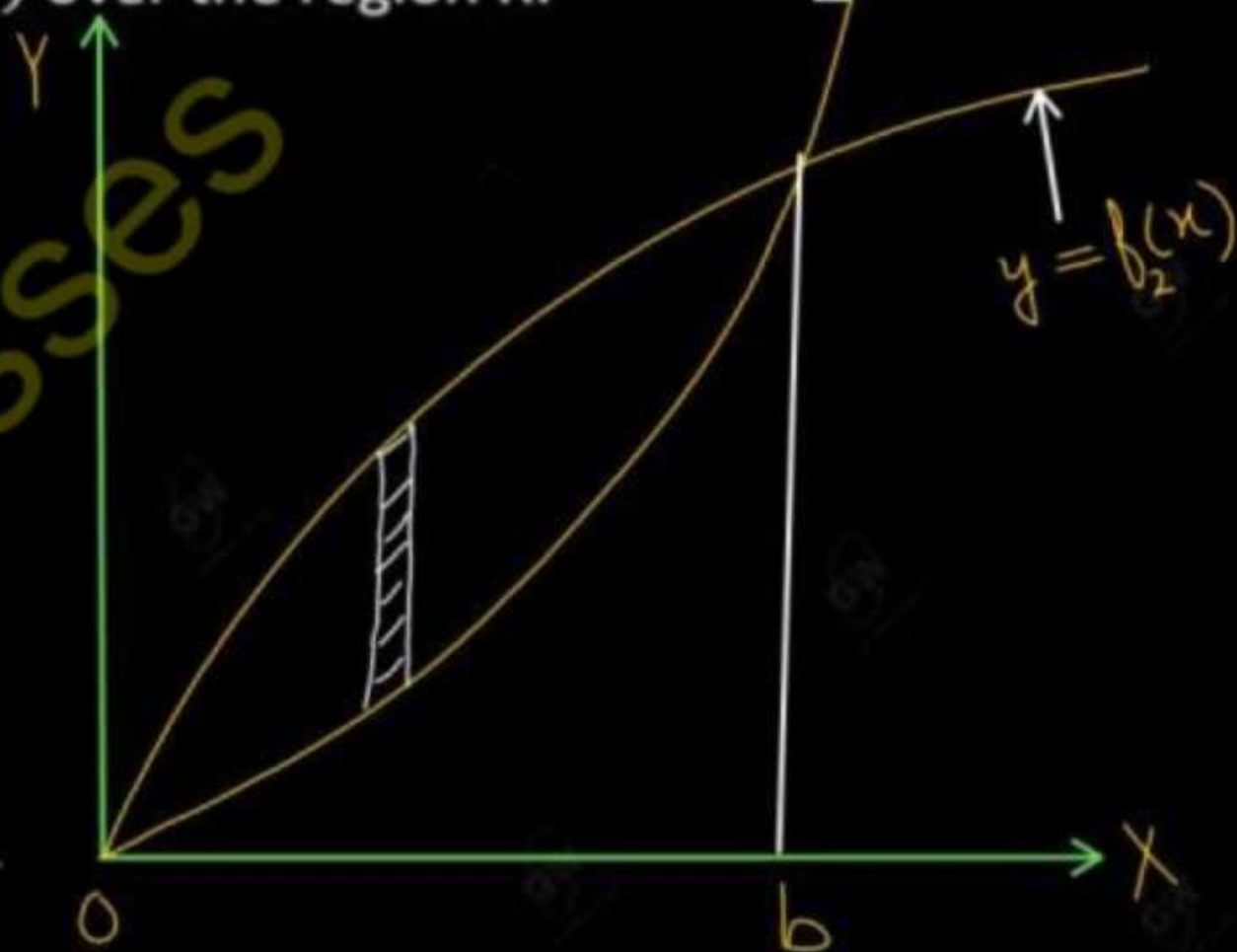
(ii)  $f_1(x) \leq y \leq f_2(x)$

$$\iint_R f(x, y) dA = \int_a^b \int_{f_1(x)}^{f_2(x)} f(x, y) dy dx$$

**Note:-** Integral with variable limits calculated first i.e. w.r.t.  $y$  (treating  $x$  as a constant)

### AREA BY DOUBLE INTEGRAL

$$A = \iint_R dA = \int_a^b \int_{f_1(x)}^{f_2(x)} dy dx$$



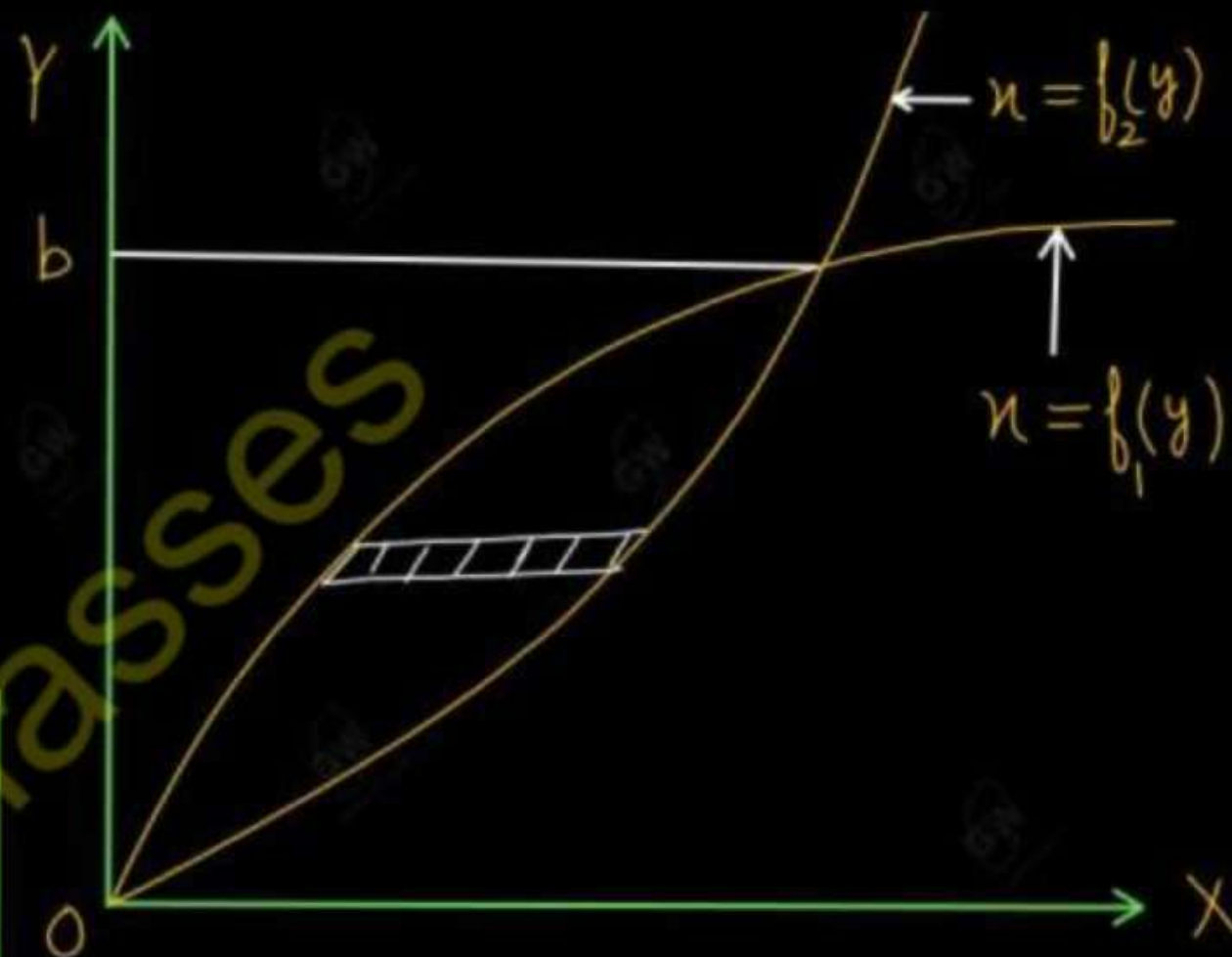


**CASE — 2 :** If the region R is defined as

(i)  $a \leq y \leq b$

(ii)  $f_1(y) \leq x \leq f_2(y)$

$$\iint_R f(x, y) dA = \int_a^b \int_{f_1(y)}^{f_2(y)} f(x, y) dx dy$$



**Note:-** Integral with variable limits calculated first i.e. w.r.t. x (treating y as a constant).

**AREA BY DOUBLE INTEGRAL**

$$A = \iint_R dA = \int_a^b \int_{f_1(y)}^{f_2(y)} dx dy$$

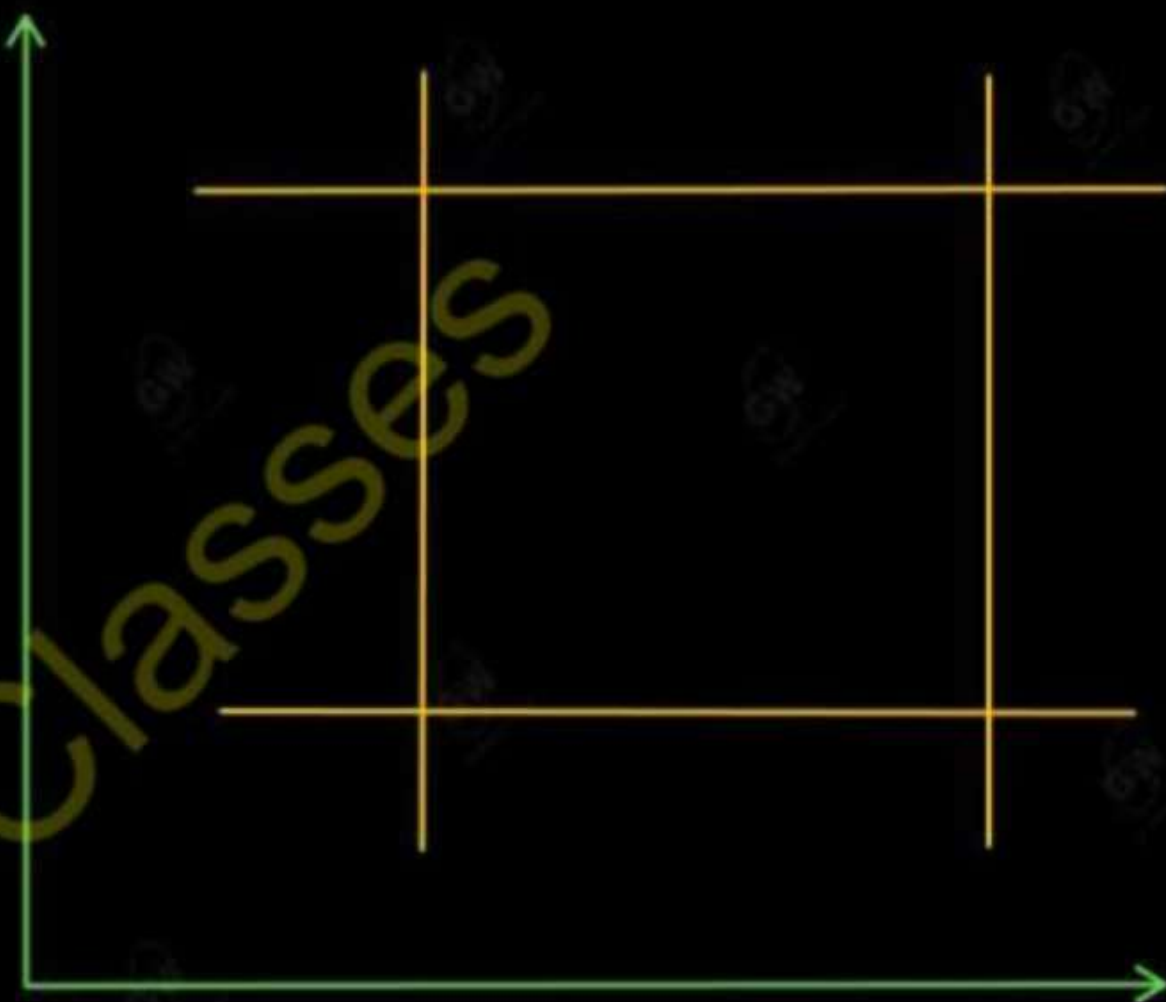


**CASE — 3** : If the region R is defined as

(i)  $a \leq x \leq b$

(ii)  $c \leq y \leq d$

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dx dy$$



Q 1 Evaluate

$$\int_0^1 \int_0^1 \frac{dx dy}{\sqrt{(1-x^2)(1-y^2)}}$$

$$\int_0^1 \int_0^1 \frac{dx dy}{\sqrt{(1-x^2)(1-y^2)}}$$

$$= \int_0^1 \int_0^1 \frac{dx dy}{(\sqrt{1-x^2})(\sqrt{1-y^2})}$$

$$= \int_0^1 \frac{dy}{\sqrt{1-y^2}} \int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

$$= \int_0^1 \frac{dy}{\sqrt{1-y^2}} (\sin^{-1} x)_0^1$$

$$= \int_0^1 \frac{dy}{\sqrt{1-y^2}} (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \int_0^1 \frac{dy}{\sqrt{1-y^2}} \left( \frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi}{2} \int_0^1 \frac{dy}{\sqrt{1-y^2}}$$

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$$= \frac{\pi}{2} (\sin^{-1} y)_0^1$$

$$= \frac{\pi}{2} (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \frac{\pi}{2} \left( \frac{\pi}{2} - 0 \right)$$

$$= \frac{\pi^2}{4}$$

$$\int_0^1 \int_0^1 \frac{dx dy}{\sqrt{(1-x^2)(1-y^2)}} = \frac{\pi^2}{4}$$



$$\int_0^1 \int_0^{x^2} x e^y dy dx$$

$$\int_0^1 \int_0^{x^2} x e^y dy dx$$

$$= \int_0^1 x dx \int_0^{x^2} e^y dy$$

$$= \int_0^1 x dx (e^y)_0^{x^2}$$

$$= \int_0^1 x dx (e^{x^2} - 1)$$

$$= \int_0^1 x (e^{x^2} - 1) dx$$

$$= \int_0^1 (x e^{x^2} - x) dx$$

$$= \int_0^1 x e^{x^2} dx - \int_0^1 x dx$$

Put  $x^2 = t$   
 $2x dx = dt$   
 $x dx = \frac{dt}{2}$

$$= \frac{1}{2} \int_0^1 e^t dt - \left( \frac{x^2}{2} \right)_0^1$$

$$= \frac{1}{2} (e^t)_0^1 - \left( \frac{1}{2} - 0 \right)$$

$$= \frac{1}{2} (e^{x^2})_0^1 - \frac{1}{2}$$

$$= \frac{1}{2} (e - 1) - \frac{1}{2}$$

$$= \frac{e}{2} - \frac{1}{2} - \frac{1}{2}$$

$$= \frac{e}{2} - 1$$

$$\int_0^1 \int_0^{x^2} x e^y dy dx = \frac{e}{2} - 1$$

Q 3

Evaluate

$$\int_0^1 \int_0^{x^2} e^{y/x} dy dx$$

$$\int_0^1 \int_0^{x^2} e^{y/x} dy dx$$

$$= \int_0^1 dx \int_0^{x^2} e^{y/x} dy$$

$$= \int_0^1 dx \int_0^{x^2} e^{\frac{1}{x}y} dy$$

$$= \int_0^1 dx \left( \frac{e^{\frac{1}{x}y}}{\frac{1}{x}} \right)_0^{x^2}$$

$$= \int_0^1 dx \left( e^{\frac{1}{x}y} \right)_0^{x^2}$$

$$= \int_0^1 dx (e^x - 1)$$

$$= \int_0^1 (xe^x - 1) dx$$

$$= \int_0^1 xe^x dx - \int_0^1 1 dx$$

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$$= \left[ (x)(e^x) - (1)(e^x) \right]_0^1 - \left( \frac{x^2}{2} \right)_0^1$$

$$= (e - e) - (0 - 1) - \frac{1}{2}(1 - 0)$$

$$= 0 + 1 - \frac{1}{2}$$

$$= \frac{1}{2}$$

$$\int_0^1 \int_0^{x^2} e^{y/x} dy dx = \frac{1}{2}$$



$$y^2 = 4x \quad \text{--- ①}$$

$$x^2 = 4y \quad \text{--- ②}$$

Solve ① and ②

$$y = \frac{x^2}{4}$$

$$\left(\frac{x^2}{4}\right)^2 = 4x$$

$$\frac{x^4}{16} = 4x$$

$$x^4 = 64x$$

$$x^4 - 64x = 0$$

$$x(x^3 - 64) = 0$$

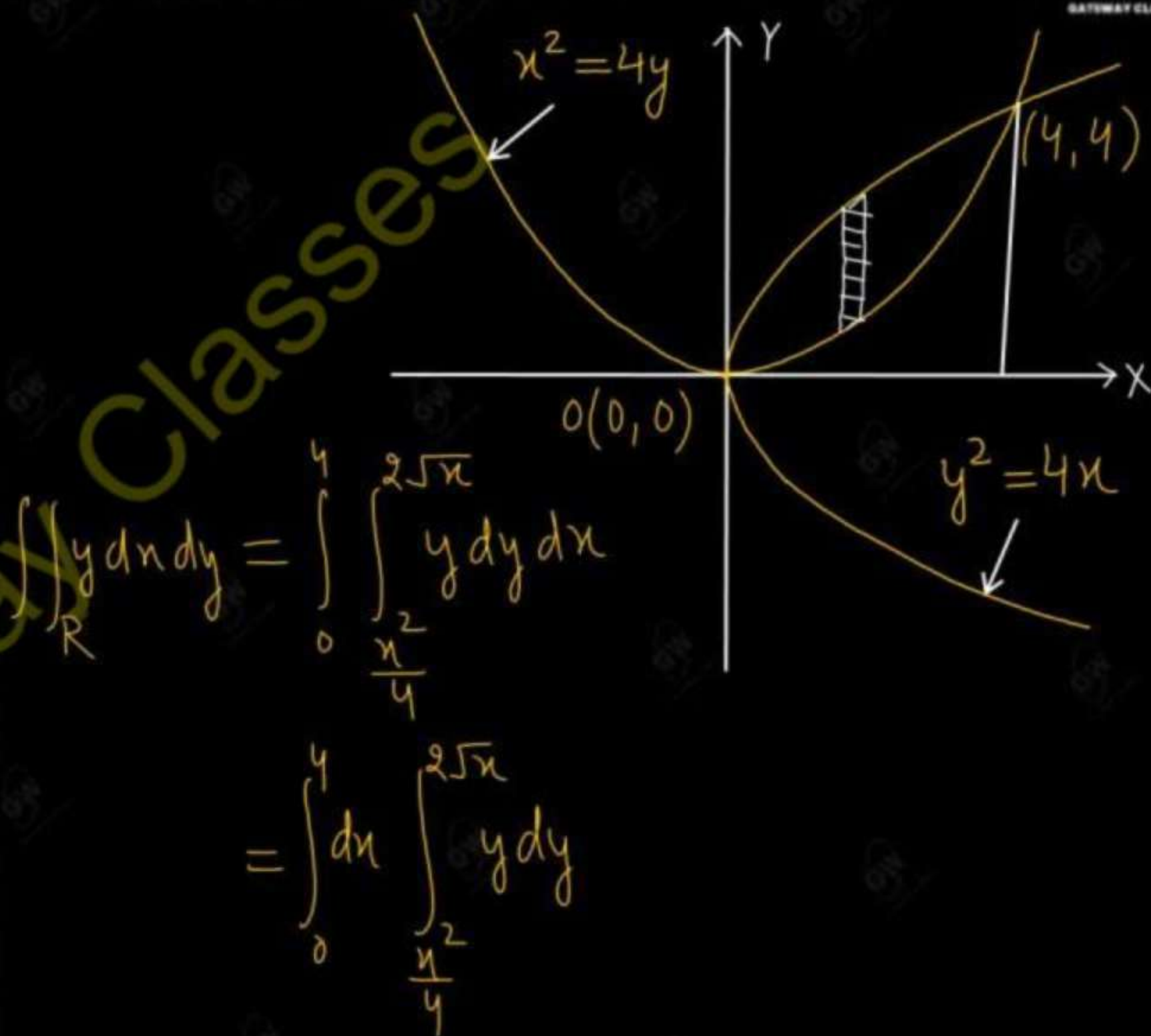
$$x = 0, 4$$

When  $x = 0$ ,  $y = 0$

When  $x = 4$ ,  $y = 4$

Point of intersection

$(0, 0)$   $(4, 4)$



$$\iint_R y \, dy \, dx = \int_0^4 dx \left( \frac{y^2}{2} \right)_{\frac{x}{4}}^{2\sqrt{x}}$$

$$= \frac{1}{2} \int_0^4 dx \left( y^2 \right)_{\frac{x}{4}}^{2\sqrt{x}}$$

$$= \frac{1}{2} \int_0^4 dx \left[ (2\sqrt{x})^2 - \left( \frac{x}{4} \right)^2 \right]$$

$$= \frac{1}{2} \int_0^4 dx \left( 4x - \frac{x^2}{16} \right)$$

$$= \frac{1}{2} \int_0^4 \left( 4x - \frac{x^2}{16} \right) dx$$

$$= \frac{1}{2} \left( \frac{4x^2}{2} - \frac{x^3}{5 \times 16} \right)_0^4$$

$$= \frac{1}{2} \left( 2 \times 4 \times 4 - \frac{4 \times 4 \times 4^3}{5 \times 16} \right) = \frac{1}{2} \left( 32 - \frac{64}{5} \right)$$

$$= \left( 16 - \frac{32}{5} \right) = \frac{80 - 32}{5}$$

$$\boxed{\iint_R y \, dx \, dy = \frac{48}{5}}$$



Q.5 Evaluate  $\iint y \, dx \, dy$  over the part of the plane bounded by the line  $y = x$  and the parabola  $y = 4x - x^2$

$$y = x \quad \text{--- (1)}$$

$$y = 4x - x^2 \quad \text{--- (2)}$$

Solve (1) and (2)

$$x = 4x - x^2$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0, 3$$

When  $x = 0, y = 0$

When  $x = 3, y = 3$

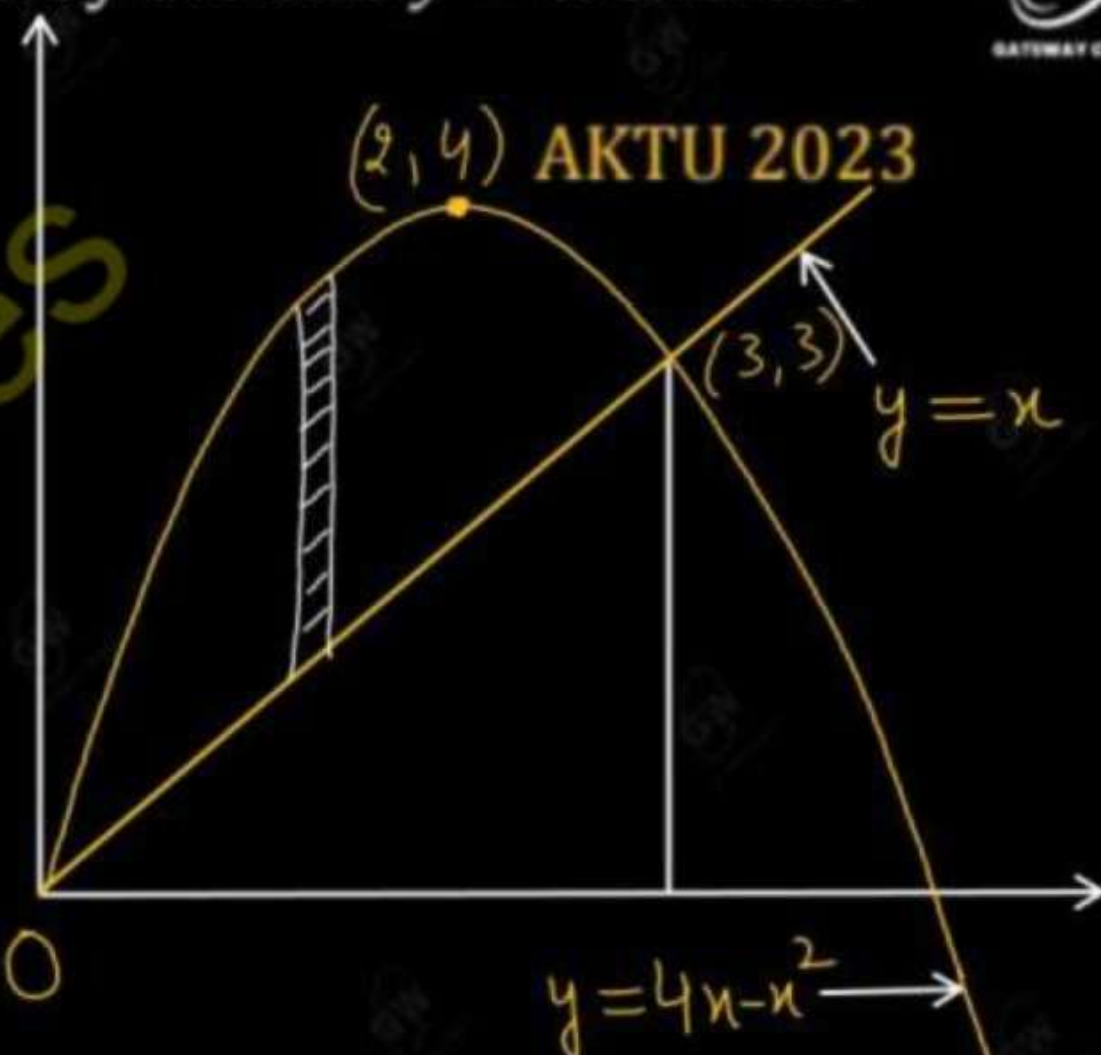
Point of intersection

$$(0, 0) (3, 3)$$

Vertex of parabola

$$y = 4x - x^2$$

$$x^2 - 4x = -y$$



$$x^2 - 4x + 4 - 4 = -y$$

$$(x-2)^2 = 4-y$$

$$(x-2)^2 = -(y-4)$$

Vertex (2, 4)

$$\iint y \, dx \, dy = \int_0^3 \int_x^{4x-x^2} y \, dy \, dx$$

$$= \int_0^3 dx \left( \frac{y^2}{2} \right)_x^{4x-x^2}$$

$$= \frac{1}{2} \int_0^3 dx \left[ (4x-x^2)^2 - x^2 \right]$$

$$= \frac{1}{2} \int_0^3 (16x^2 + x^4 - 8x^3 - x^2) \, dx$$

$$= \frac{1}{2} \int_0^3 (15x^2 + x^4 - 8x^3) \, dx$$

$$= \frac{1}{2} \left( 5x^3 + \frac{x^5}{5} - 2x^4 \right)_0^3$$

$$= \frac{1}{2} \left( 5 \times 27 + \frac{3^5}{5} - 2 \times 3^4 \right)$$

$$= \frac{1}{2} \left( 135 + \frac{243}{5} - 162 \right)$$

$$= \frac{1}{2} \left( \frac{675 + 243 - 810}{5} \right)$$

$$= \frac{1}{2} \times \frac{108}{5}$$

$$= \frac{54}{5}$$



Q.6 Evaluate  $\iint xy \, dx \, dy$ , over the positive quadrant of the circle  $x^2 + y^2 = a^2$

$$x^2 + y^2 = a^2$$

$$y^2 = a^2 - x^2$$

$$y = \sqrt{a^2 - x^2}$$

$$\iint xy \, dx \, dy = \int_0^a \int_0^{\sqrt{a^2 - x^2}} xy \, dy \, dx$$

$$= \int_0^a x \, dx \int_0^{\sqrt{a^2 - x^2}} y \, dy$$

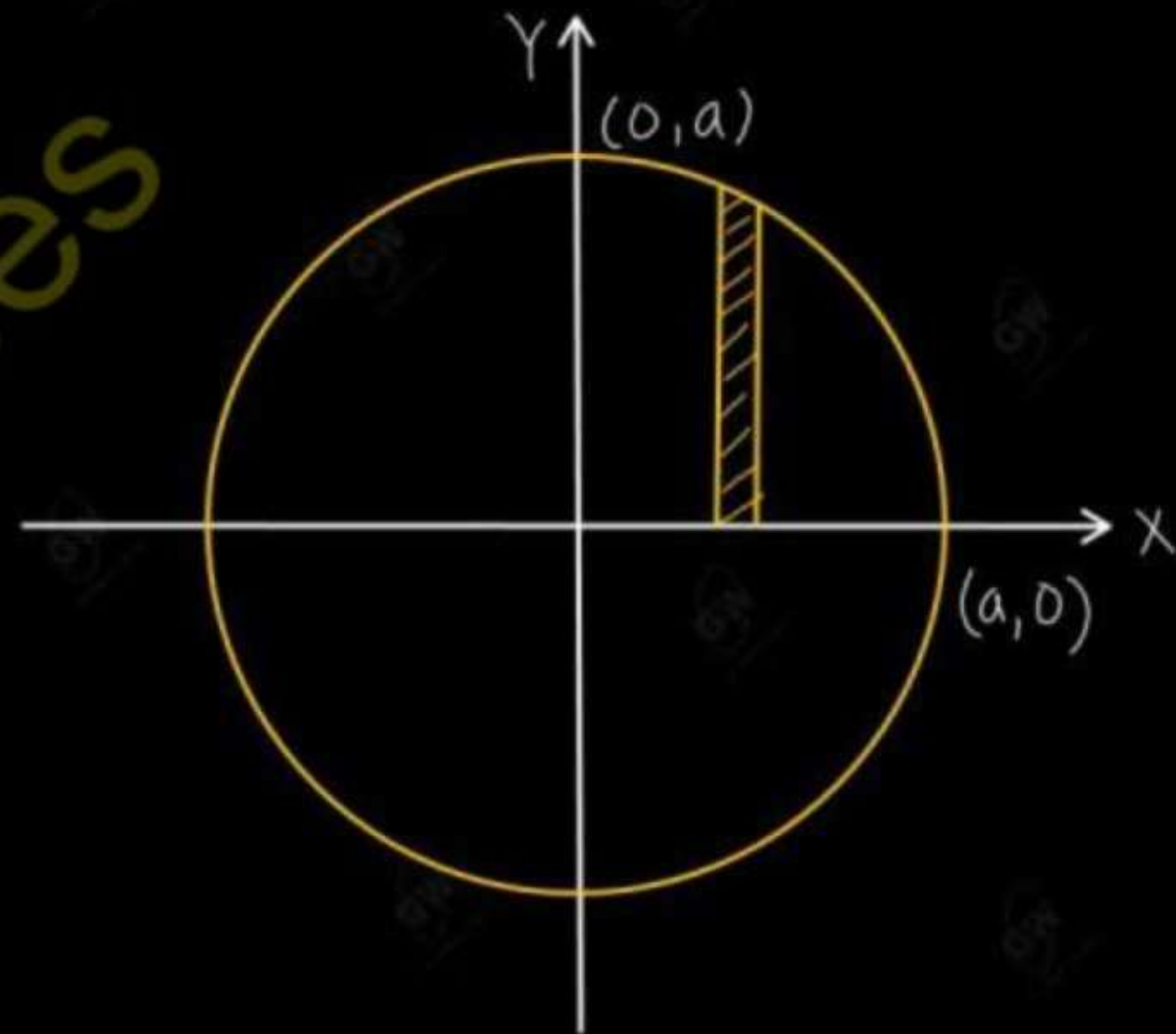
$$= \int_0^a x \left( \frac{y^2}{2} \right)_0^{\sqrt{a^2 - x^2}}$$

$$= \int_0^a x \left( \frac{a^2 - x^2}{2} \right) dx$$

$$= \frac{1}{2} \int_0^a (a^2 x - x^3) dx$$

$$= \frac{1}{2} \left[ \frac{a^2 x^2}{2} - \frac{x^4}{4} \right]_0^a$$

$$= \frac{1}{2} \left[ \frac{a^4}{2} - \frac{a^4}{4} \right] = \frac{1}{2} \left( \frac{2a^4 - a^4}{4} \right) = \boxed{\frac{a^4}{8}}$$



Q.7 Evaluate  $\iint xy \, dx \, dy$ , over the region in positive quadrant for which  $x + y \leq 1$

$$x + y \leq 1$$

$$x + y = 1$$

|   |   |   |
|---|---|---|
| x | 0 | 1 |
| y | 1 | 0 |

$$y = 1 - x$$

$$\iint xy \, dx \, dy = \int_0^1 \int_0^{1-x} xy \, dx \, dy$$

$$= \int_0^1 x \, dx \int_0^{1-x} y \, dy$$

$$= \int_0^1 x \, dx \left( \frac{y^2}{2} \right)_0^{1-x}$$

$$= \frac{1}{2} \int_0^1 x(1-x)^2 \, dx$$

$$= \frac{1}{2} \int_0^1 x(1+x^2-2x) \, dx$$

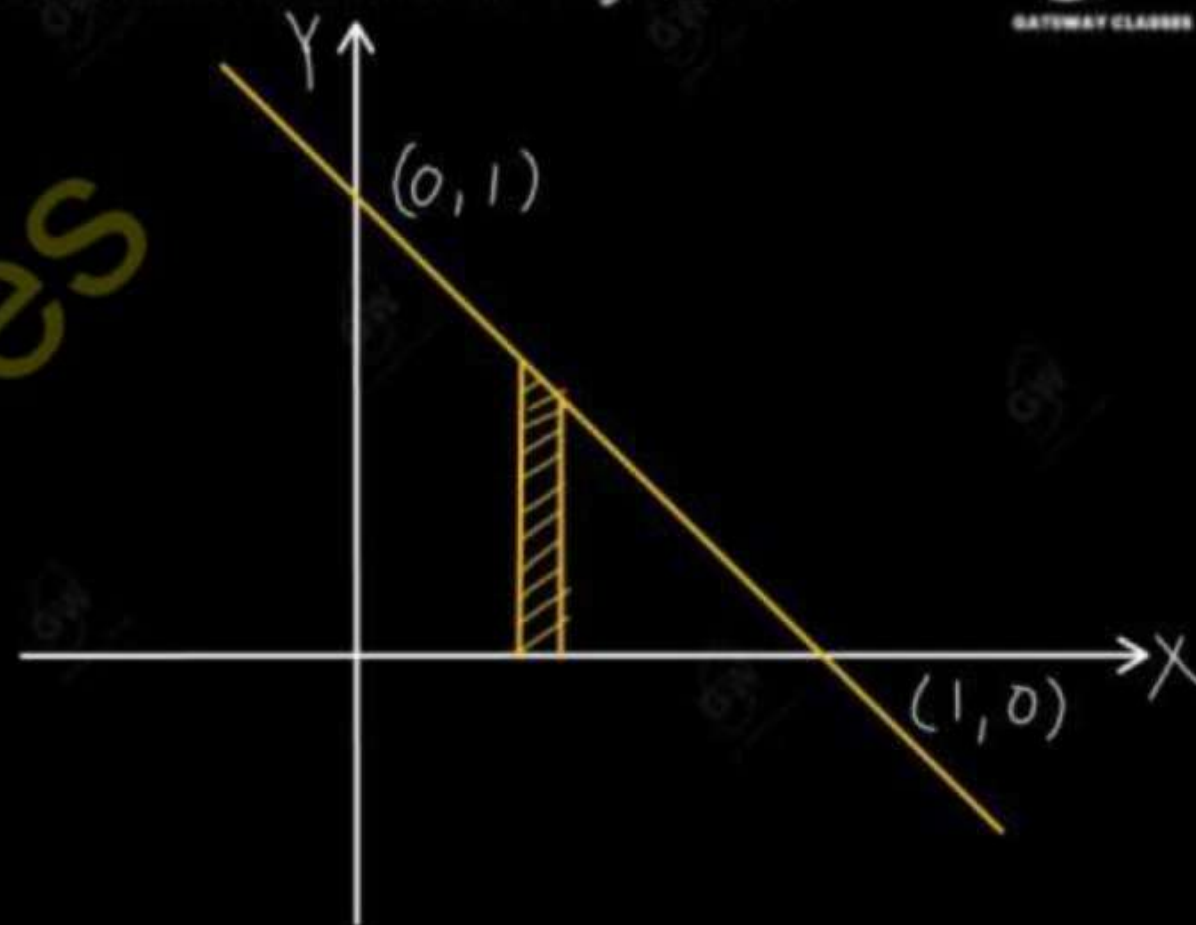
$$= \frac{1}{2} \int_0^1 (x+x^3-2x^2) \, dx$$

$$= \frac{1}{2} \left( \frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{3} \right)_0^1$$

$$= \frac{1}{2} \left( \frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right)$$

$$= \frac{1}{2} \left( \frac{6+3-8}{12} \right)$$

$$= \frac{1}{24}$$





$$y^2 = 4ax \quad \text{--- ①}$$

$$x^2 = 4ay \quad \text{--- ②}$$

$$y = \frac{x^2}{4a}$$

Put  $y$  in ①

$$\left(\frac{x^2}{4a}\right)^2 = 4ax$$

$$\frac{x^4}{16a^2} = 4ax$$

$$x^4 = 64a^3x$$

$$x^4 - 64a^3x = 0$$

$$x(x^3 - 64a^3) = 0$$

$$x = 0$$

$$x = 4a$$

When  $x = 0, y = 0$

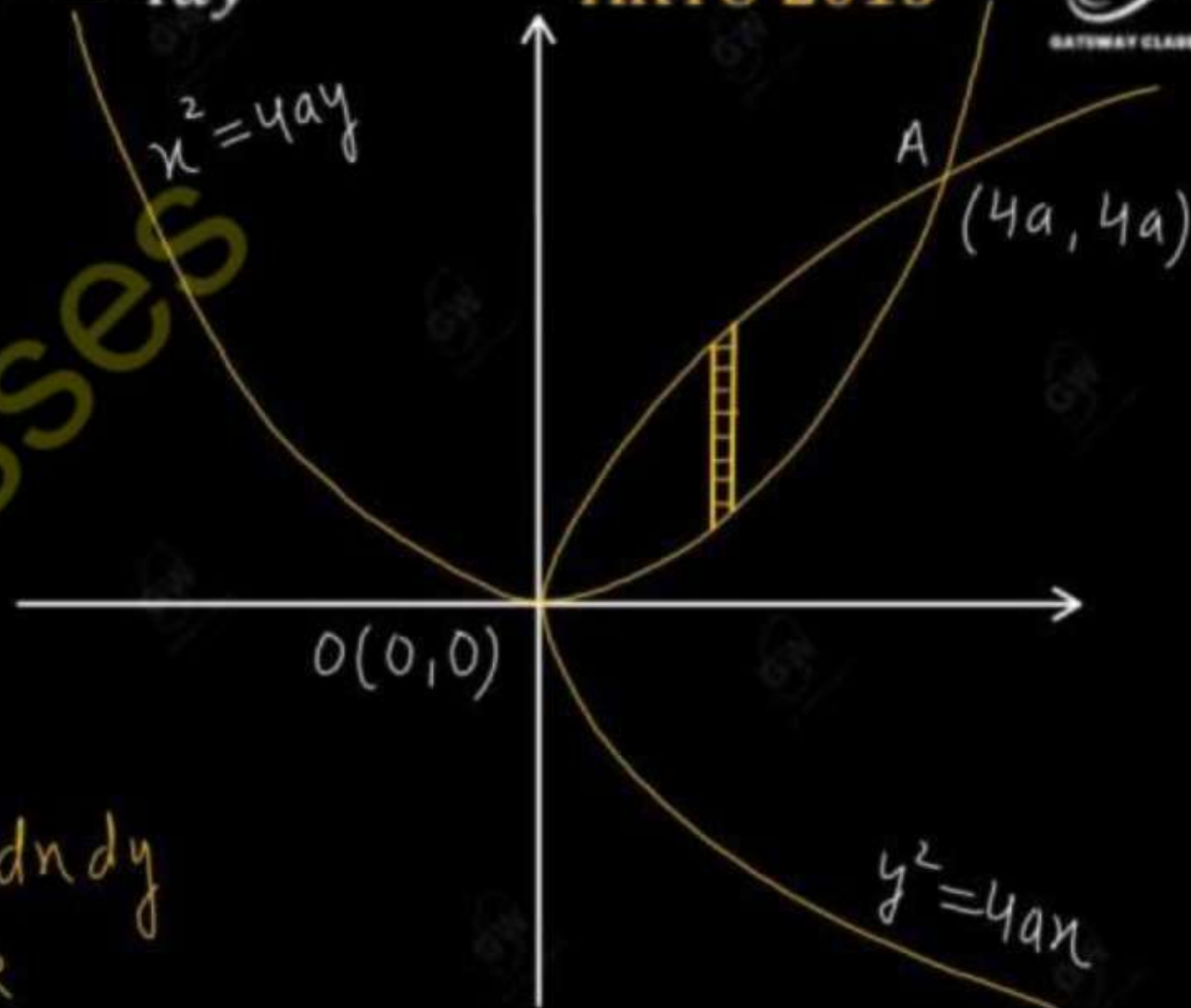
When  $x = 4a, y = 4a$

Point of intersection

$O(0,0)$  and  $A(4a,4a)$

$$A = \int \int_R dx dy$$

$$A = \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dx dy$$



$$A = \int_0^{4a} dx \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy$$

$$= \int_0^{4a} dx \left( y \right)_{\frac{x^2}{4a}}^{2\sqrt{ax}}$$

$$= \int_0^{4a} \left( 2\sqrt{ax} - \frac{x^2}{4a} \right) dx$$

$$= \int_0^{4a} \left( 2\sqrt{a} x^{\frac{1}{2}} - \frac{x^2}{4a} \right) dx$$

$$= \left( 2\sqrt{a} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{3 \times 4a} \right)_0^{4a}$$

$$= \frac{4\sqrt{a}}{3} (4a)^{\frac{3}{2}} - \frac{(4a)^3}{12a}$$

$$= \frac{4\sqrt{a}}{3} (2\sqrt{a})^3 - \frac{\cancel{4a} \times 4a \times 4a}{\cancel{12a} \cdot 3}$$

$$= \frac{4\sqrt{a} \times 8a\sqrt{a}}{3} - \frac{16a^2}{3}$$

$$= \frac{32a^2 - 16a^2}{3}$$

$$A = \frac{16}{3}a^2$$



UNIT -4 : DPP - 1

Topic : Double Integral

Q.1 Evaluate  $\int_0^2 \int_0^1 (x^2 + 3y^2) dy dx$

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Q.2 Evaluate  $\int_0^a \int_0^x xy dy dx$

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Q.3 Evaluate  $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dx dy}{1+x^2+y^2}$

Q.4 Evaluate  $\int_0^2 \int_{y^2}^y (1 + xy^2) dx dy$

Q.5 Evaluate  $\iint e^{2x+3y} dx dy$ , over the triangle bounded by  $x = 0$ ,  $y = 0$  and  $x + y = 1$

Q.6 Find the area bounded by the curve  $y^2 = x$  and  $x^2 = y$

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## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 2

#### Today's target

- CHANGE OF ORDER OF INTEGRATION (PART – 1)
- PYQs
- DPP



By Gulshan sir



- In a double integral with variable limits, the change of order of integration changes the limits of integration

- Some complicated problems becomes easy after changing the order of integration

## Working Rule

**STEP – 1 :** Write the given limits of integration

**STEP – 2 :** Draw the region for a given limits of integration

**STEP – 3 :** Check the strip for variable limits in the given region

**STEP – 4 :** If the given strip is horizontal then take a new vertical strip

OR

If the given strip is vertical then take a new horizontal strip

**STEP – 5 :** Change the limit of integration, according to new strip

CASE - 1  $\begin{cases} X \rightarrow C \\ Y \rightarrow V \end{cases}$

CASE - 2  $\begin{cases} X \rightarrow V \\ Y \rightarrow C \end{cases}$

Q.1 Evaluate the integrals by changing the order of integration  $\int_0^4 \int_y^4 \frac{x dx dy}{x^2 + y^2}$

$$\int_0^4 \int_y^4 \frac{x dx dy}{x^2 + y^2}$$

Given limits of integration

$$y = 0 \checkmark$$

$$y = 4 \checkmark$$

$$x = y \checkmark$$

$$x = 4 \checkmark$$

Take a new vertical strip

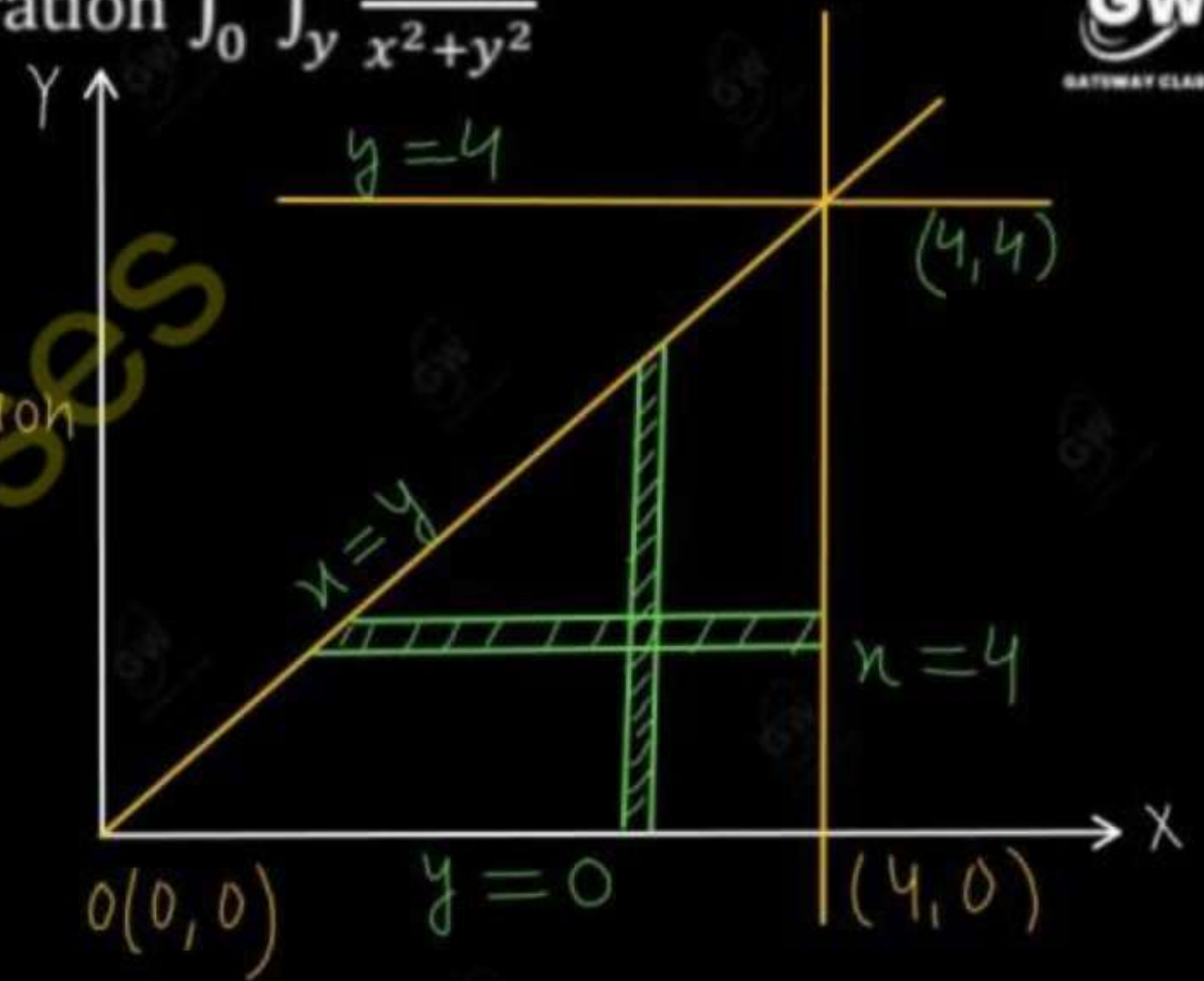
New limits of integration

$$x = 0$$

$$x = 4$$

$$y = 0$$

$$y = x$$



change the order of integration

$$\int_0^4 \int_y^4 \frac{x dx dy}{x^2 + y^2} = \int_0^4 \int_0^x \frac{x dx dy}{x^2 + y^2}$$

$\Rightarrow$  Given strip is horizontal



$$\int_0^4 \int_y^4 \frac{x \, dx \, dy}{x^2 + y^2} = \int_0^4 x \, dx \int_0^x \frac{dy}{x^2 + y^2}$$

$$= \int_0^4 x \, dx \left( \frac{1}{x} \tan^{-1} \frac{y}{x} \right)_0^x$$

$$= \int_0^4 \cancel{x} \times \frac{1}{\cancel{x}} \, dx \left( \tan^{-1} \frac{y}{x} \right)_0^x$$

$$= \int_0^4 dx \left( \tan^{-1} 1 - \tan^{-1} 0 \right)$$

$$= \int_0^4 \left( \frac{\pi}{4} - 0 \right) dx$$

$$= \frac{\pi}{4} \int_0^4 dx$$

$$= \frac{\pi}{4} (x)_0^4$$

$$= \frac{\pi}{4} \times 4$$

$$\int_0^4 \int_y^4 \frac{x \, dx \, dy}{x^2 + y^2} = \pi$$

Q. 2 Change the order of integration in  $I = \int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$  and hence evaluate the same

$$I = \int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$$

Given limits of integration

$$x = 0 \checkmark$$

$$x = 1 \checkmark$$

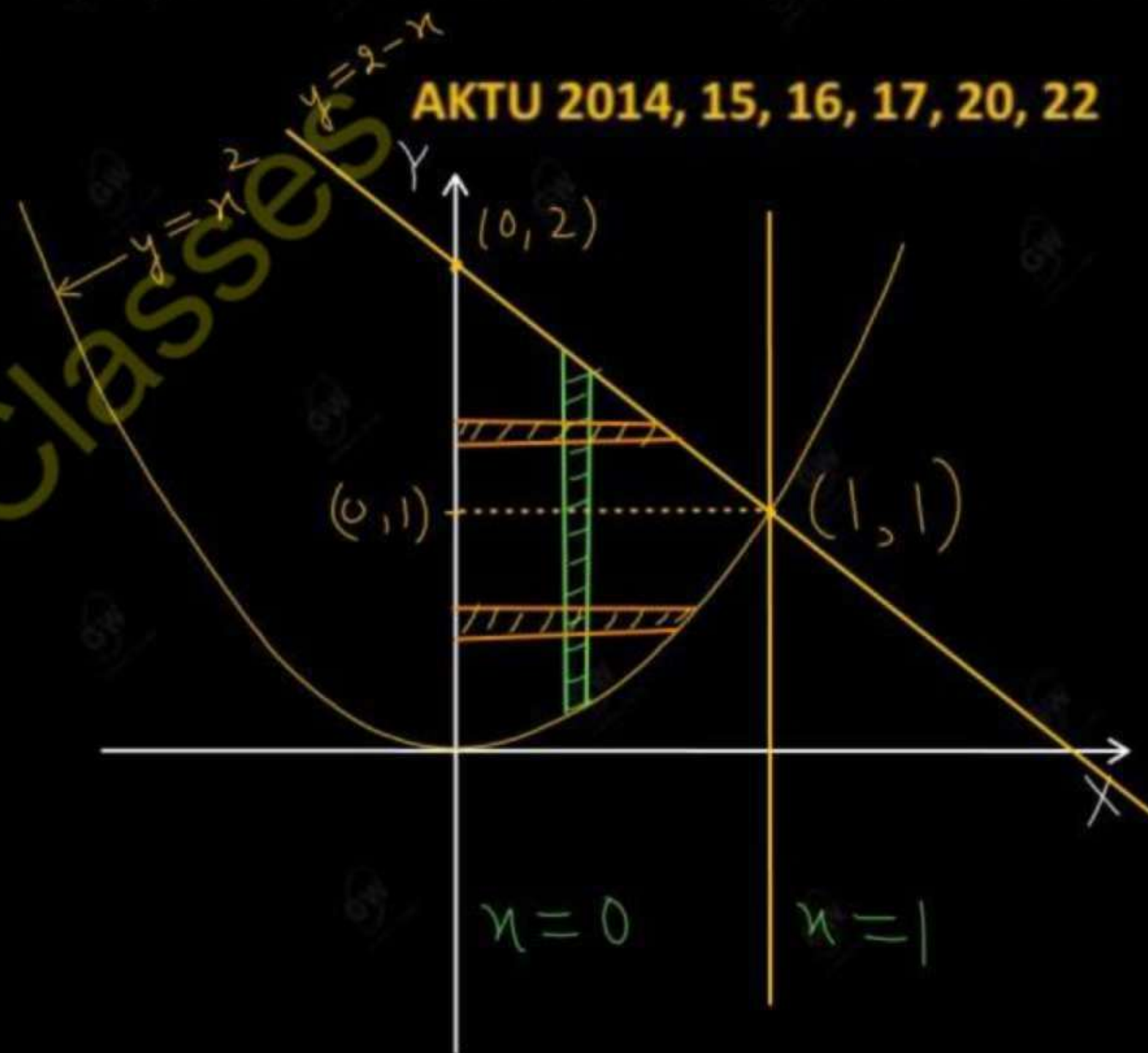
$$y = x^2 \checkmark$$

$$y = 2-x \checkmark$$

|   |   |   |
|---|---|---|
| x | 0 | 1 |
| y | 2 | 1 |

$\Rightarrow$  Given strip is vertical

Take two new vertical strip





New limits of integration

For lower region

$$y = 0$$

$$y = 1$$

$$x = 0$$

$$x = \sqrt{y}$$

For upper region

$$y = 1$$

$$y = 2$$

$$x = 0$$

$$x = 2 - y$$

Change the order of integration

$$\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx = \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy$$

$$= \int_0^1 y \, dy \int_0^{\sqrt{y}} x \, dx + \int_1^2 y \, dy \int_0^{2-y} x \, dx$$

$$= \int_0^1 y \, dy \left( \frac{x^2}{2} \right)_0^{\sqrt{y}} + \int_1^2 y \, dy \left( \frac{x^2}{2} \right)_0^{2-y}$$

$$= \int_0^1 y \, dy \times \frac{y}{2} + \int_1^2 y \, dy \frac{(2-y)^2}{2}$$

$$= \frac{1}{2} \int_0^1 y^2 dy + \frac{1}{2} \int_1^2 y(4 + y^2 - 4y) dy$$

$$= \frac{1}{2} \left( \frac{y^3}{3} \right)_0^1 + \frac{1}{2} \int_1^2 (4y + y^3 - 4y^2) dy$$

$$= \frac{1}{2} \left( \frac{1}{3} - 0 \right) + \frac{1}{2} \left( 2y^2 + \frac{y^4}{4} - \frac{4y^3}{3} \right)_1^2$$

$$= \frac{1}{6} + \frac{1}{2} \left[ 8 + 4 - \frac{32}{3} - \left( 2 + \frac{1}{4} - \frac{4}{3} \right) \right]$$

$$= \frac{1}{6} + \frac{1}{2} \left( 12 - \frac{32}{3} - 2 - \frac{1}{4} + \frac{4}{3} \right)$$

$$= \frac{1}{6} + \frac{1}{2} \left( 10 - \frac{32}{3} - \frac{1}{4} + \frac{4}{3} \right)$$

$$= \frac{1}{6} + \frac{1}{2} \left( \frac{120 - 128 - 3 + 16}{12} \right)$$

$$= \frac{1}{6} + \frac{1}{2} \times \frac{5}{12}$$

$$= \frac{4 + 5}{24} = \frac{9}{24} = \frac{3}{8}$$

$$I = \frac{3}{8}$$



Change the order of integration in the following integral and evaluate  $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy dx$

$$\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy dx$$

Given limits of integration

$$x = 0$$

$$x = 4a$$

$$y = \frac{x^2}{4a} \Rightarrow x^2 = 4ay$$

$$y = 2\sqrt{ax} \Rightarrow y^2 = 4ax$$

Given strip is vertical

Take a new horizontal strip

New limits of integration

$$y = 0$$

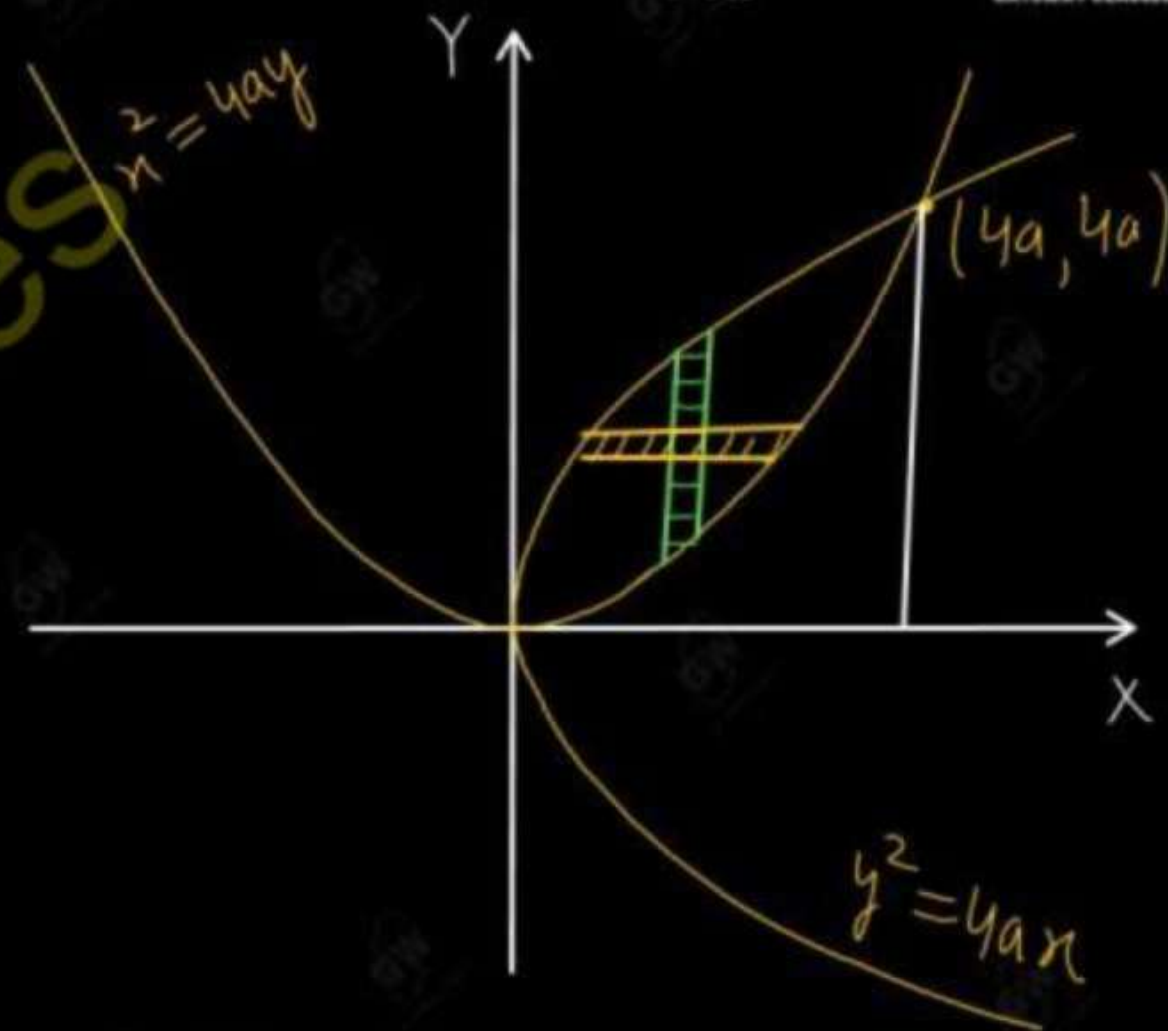
$$y = 4a$$

$$x = \frac{y^2}{4a}$$

$$x = 2\sqrt{ay}$$

Change the order of integration

$$\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy dx = \int_0^{4a} \int_{y^2/4a}^{2\sqrt{ay}} dx dy$$



$$\int_0^{4a} \int_{\frac{n^2}{4a}}^{2\sqrt{an}} dy dn = \int_0^{4a} dy \left( n \right)_{\frac{y^2}{4a}}^{2\sqrt{ay}}$$

$$= \int_0^{4a} dy \left( 2\sqrt{ay} - \frac{y^2}{4a} \right)$$

$$= \int_0^{4a} \left( 2\sqrt{a} y^{1/2} - \frac{y^2}{4a} \right) dy$$

$$= 2\sqrt{a} \int_0^{4a} y^{1/2} dy - \frac{1}{4a} \int_0^{4a} y^2 dy$$

$$= 2\sqrt{a} \times \left( \frac{y^{3/2}}{3/2} \right)_0^{4a} - \frac{1}{4a} \left( \frac{y^3}{3} \right)_0^{4a}$$

$$= \frac{4\sqrt{a}}{3} \left( y^{3/2} \right)_0^{4a} - \frac{1}{12a} \left( y^3 \right)_0^{4a}$$

$$= \frac{4\sqrt{a}}{3} \left[ (2\sqrt{a})^3 - 0 \right] - \frac{1}{12a} (4a)^3$$

$$= \frac{4\sqrt{a}}{3} \times 8a\sqrt{a} - \frac{1}{12a} \times 4 \times 4 \times 4a^3$$

$$= \frac{32a^2}{3} - \frac{16a^2}{3}$$

$$= \frac{16}{3} a^2$$

$$\int_0^{4a} \int_{\frac{n^2}{4a}}^{2\sqrt{an}} dy dn = \frac{16a^2}{3}$$



$$\int_0^a \int_{\sqrt{ax}}^a \frac{y^2 dy dx}{\sqrt{y^4 - a^2 x^2}}$$

$$\int_0^a \int_{\sqrt{ax}}^a \frac{y^2 dy dx}{\sqrt{y^4 - a^2 x^2}}$$

Given limits of integration

$$x = 0 \quad \checkmark$$

$$x = a \quad \checkmark$$

$$y = \sqrt{ax} \Rightarrow y^2 = ax$$

$$y = a \quad \checkmark$$

$\Rightarrow$  Given strip is vertical

Take a new Horizontal strip

New limits of integration

$$y = 0$$

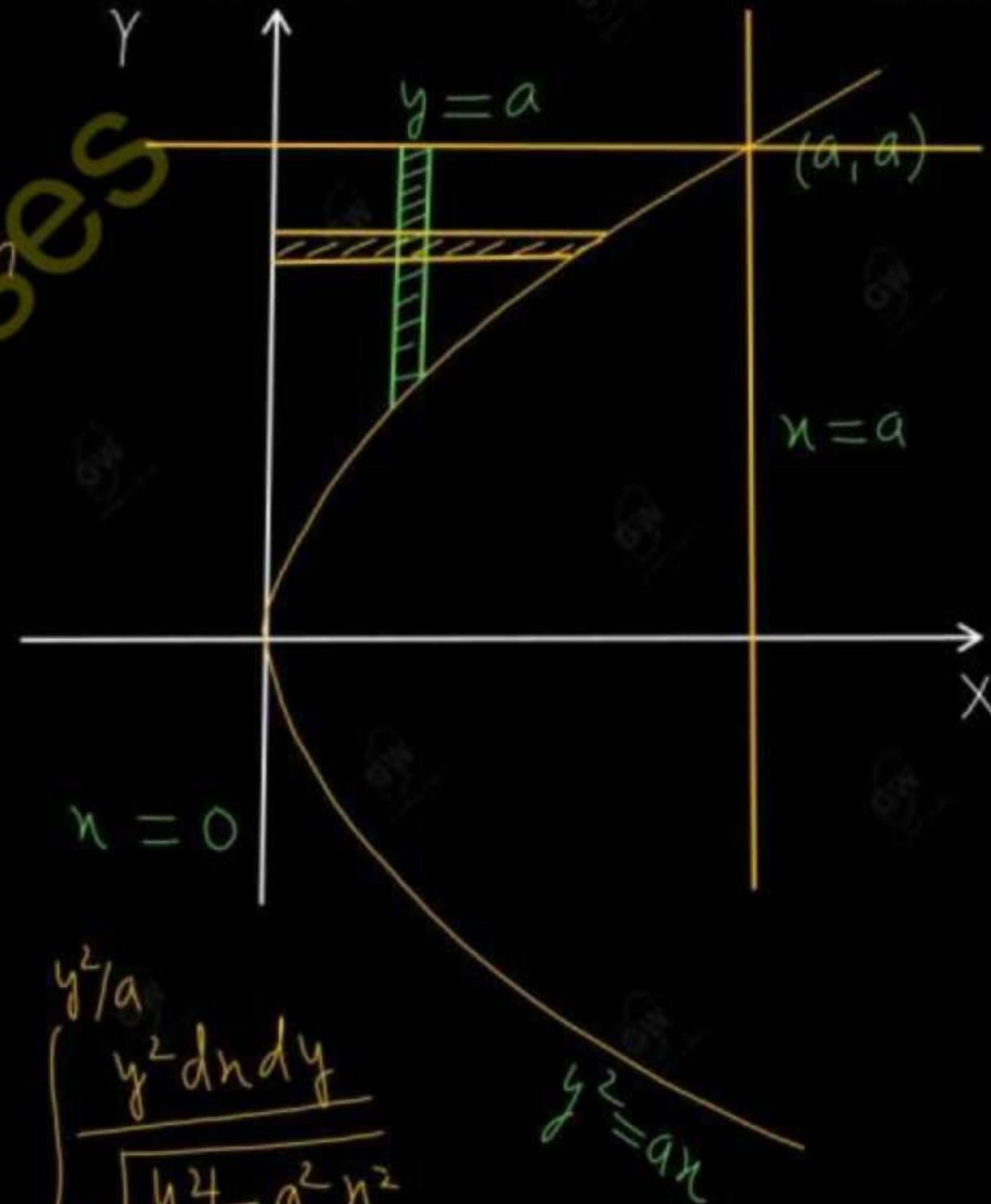
$$y = a$$

$$x = 0$$

$$x = y^2/a$$

Change the order of integration

$$\int_0^a \int_{\sqrt{ax}}^a \frac{y^2 dy dx}{\sqrt{y^4 - a^2 x^2}} = \int_0^a \int_0^{y^2/a} \frac{y^2 dx dy}{\sqrt{y^4 - a^2 x^2}}$$



$$\begin{aligned} \int_0^a \int_{\sqrt{an}}^a \frac{y^2 dy dn}{\sqrt{y^4 - a^2 n^2}} &= \int_0^a y^2 dy \int_0^{y^2/a} \frac{dn}{\sqrt{y^4 - a^2 n^2}} \\ &= \frac{1}{a} \int_0^a y^2 dy \int_0^{y^2/a} \frac{dn}{\sqrt{\left(\frac{y^2}{a}\right)^2 - n^2}} \\ &= \frac{1}{a} \int_0^a y^2 dy \left( \sin^{-1} \frac{n}{\frac{y^2}{a}} \right) \Big|_0^{y^2/a} \\ &= \frac{1}{a} \int_0^a y^2 dy \left( \sin^{-1} \frac{an}{y^2} \right) \Big|_0^{y^2/a} \end{aligned}$$

$$= \frac{1}{a} \int_0^a y^2 dy \left( \sin^{-1} 1 - \sin^{-1} 0 \right)$$

$$= \frac{\pi}{2a} \int_0^a y^2 dy = \frac{\pi}{2a} \left( \frac{y^3}{3} \right) \Big|_0^a$$

$$= \frac{\pi}{2a \times 3} (a^3 - 0)$$

$$= \frac{\pi a^2}{6}$$

$$\int_0^a \int_{\sqrt{an}}^a \frac{y^2 dy dn}{\sqrt{y^4 - a^2 n^2}} = \frac{\pi a^2}{6}$$



**GW** Q.5 Change the order of integration in  $\int_0^1 \int_{x^2}^{2-x} f(x, y) dy dx$

**AKTU 2018**

**GW**  
GATEWAY CLASSES

Same as Q2

Gateway Classes

UNIT -4 : DPP - 2

TOPIC : CHANGE OF ORDER OF INTEGRATION

Q.1 Evaluate the integrals by changing the order of integration  $\int_0^a \int_y^a \frac{xdxdy}{x^2+y^2}$

Q.2 Evaluate the integrals by changing the order of integration  $\int_0^{2a} \int_{x^2/4a}^{3a-x} (x^2 + y^2) dy dx$

Q.3 Evaluate the integrals by changing the order of integration  $\int_0^2 \int_{x^2/4}^{3-x} xydy dx$

AKTU 2018, 19

Q.4 Changing the order of integration in the double integral  $I = \int_0^8 \int_{x/4}^2 f(x,y)dy dx$  leads to

the value  $I = \int_r^s \int_p^q f(x,y)dx dy$ . What is the value of q?

AKTU 2017



## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 3

#### Today's target

- CHANGE OF ORDER OF INTEGRATION (PART – 2)
- PYQs
- DPP



By Gulshan sir



**GW** Q. 1 Evaluate the integrals by changing the order of integration

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dy dx$$

Given limits of integration

$$x=0 \checkmark$$

$$x=1 \checkmark$$

$$y=0 \checkmark$$

$$y=\sqrt{1-x^2}$$

$$\Rightarrow y^2 = 1-x^2$$

$$\Rightarrow \boxed{x^2 + y^2 = 1} \checkmark$$

$\Rightarrow$  Given strip is vertical

Take a new Horizontal Strip

New limits of integration

$$y=0$$

$$y=1$$

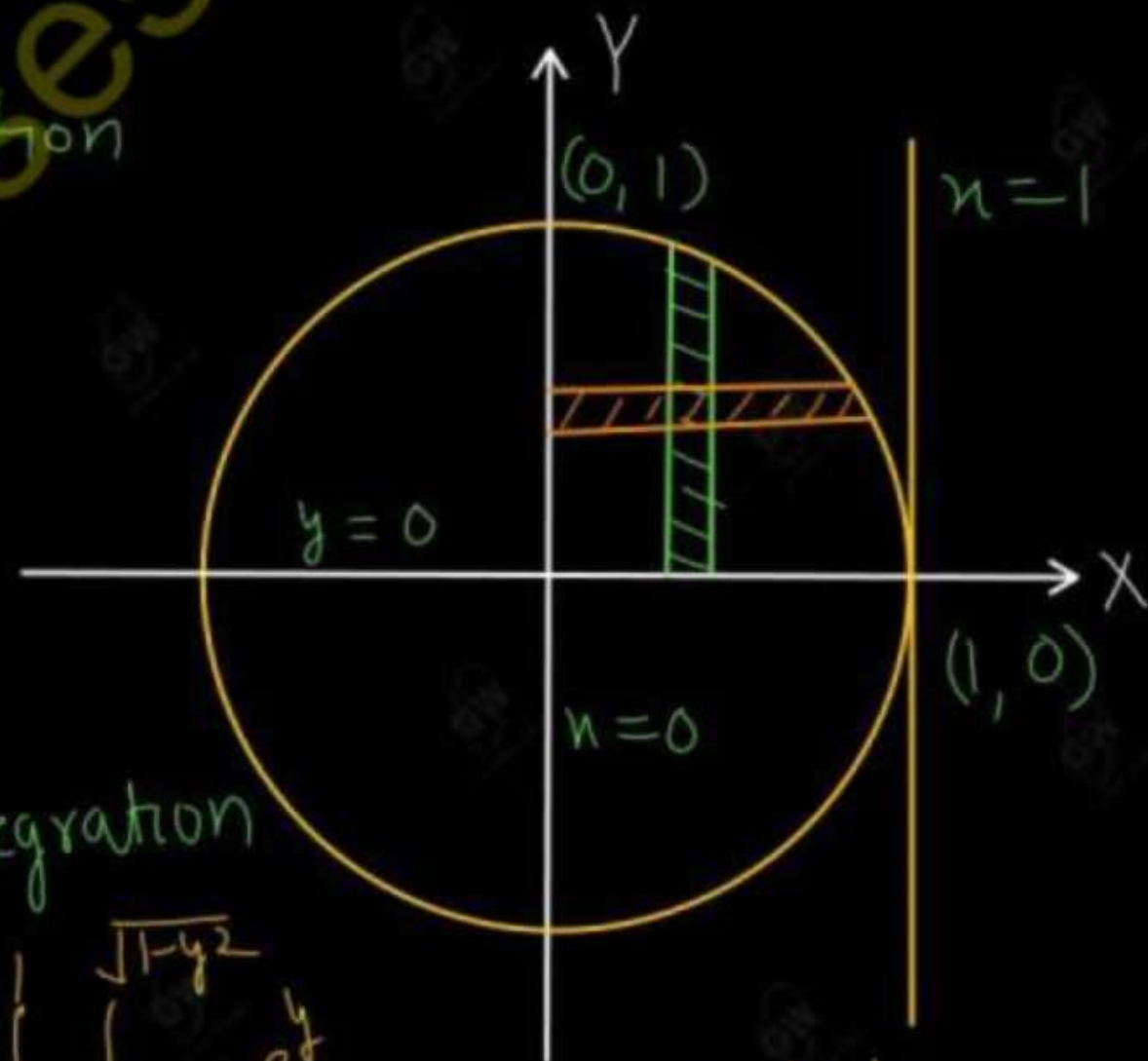
$$x=0$$

$$x=\sqrt{1-y^2}$$

Change the order of integration

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dy dx =$$

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dx dy$$





$$= \int_0^1 \frac{e^y dy}{(e^y + 1)} \int_0^{\sqrt{1-y^2}} \frac{dn}{\sqrt{1-n^2-y^2}}$$

$$= \int_0^1 \frac{e^y dy}{(e^y + 1)} \int_0^{\sqrt{1-y^2}} \frac{dn}{\sqrt{(\sqrt{1-y^2})^2 - n^2}}$$

$$= \int_0^1 \frac{e^y}{e^y + 1} dy \left( \sin^{-1} \frac{n}{\sqrt{1-y^2}} \right)_0^{\sqrt{1-y^2}}$$

$$= \int_0^1 \frac{e^y}{e^y + 1} dy (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \frac{\pi}{2} \int_0^1 \frac{e^y}{e^y + 1} dy$$

$$= \frac{\pi}{2} \left[ \log(e^y + 1) \right]_0^1$$

$$= \frac{\pi}{2} [\log(e+1) - \log 2]$$

$$= \frac{\pi}{2} \log\left(\frac{e+1}{2}\right)$$

$$\int_0^1 \int_0^{\sqrt{1-n^2}} \frac{e^y}{(e^y + 1)\sqrt{1-n^2-y^2}} dy dn = \frac{\pi}{2} \log\left(\frac{e+1}{2}\right)$$

GW  
GATEWAY CLASSES

$$\int \frac{f'(x)}{f(x)} dx = \log f(x)$$

Q. 2 Evaluate the integrals by changing the order of integration

$$\int_0^1 \int_{e^x}^e \frac{dy dx}{\log y}$$

$$\int_0^1 \int_{e^x}^e \frac{dy dx}{\log y}$$

Given limits of integration

$$x = 0 \text{ --- (1)}$$

$$x = 1 \text{ --- (2)}$$

$$y = e^x \text{ --- (3)}$$

$$y = e \text{ --- (4)}$$

Solve (3) and (4)

$$e^x = e$$

$$x = 1$$

⇒ Given strip is vertical.

Take a new Horizontal strip

New limits of integration

$$y = 1$$

$$y = e$$

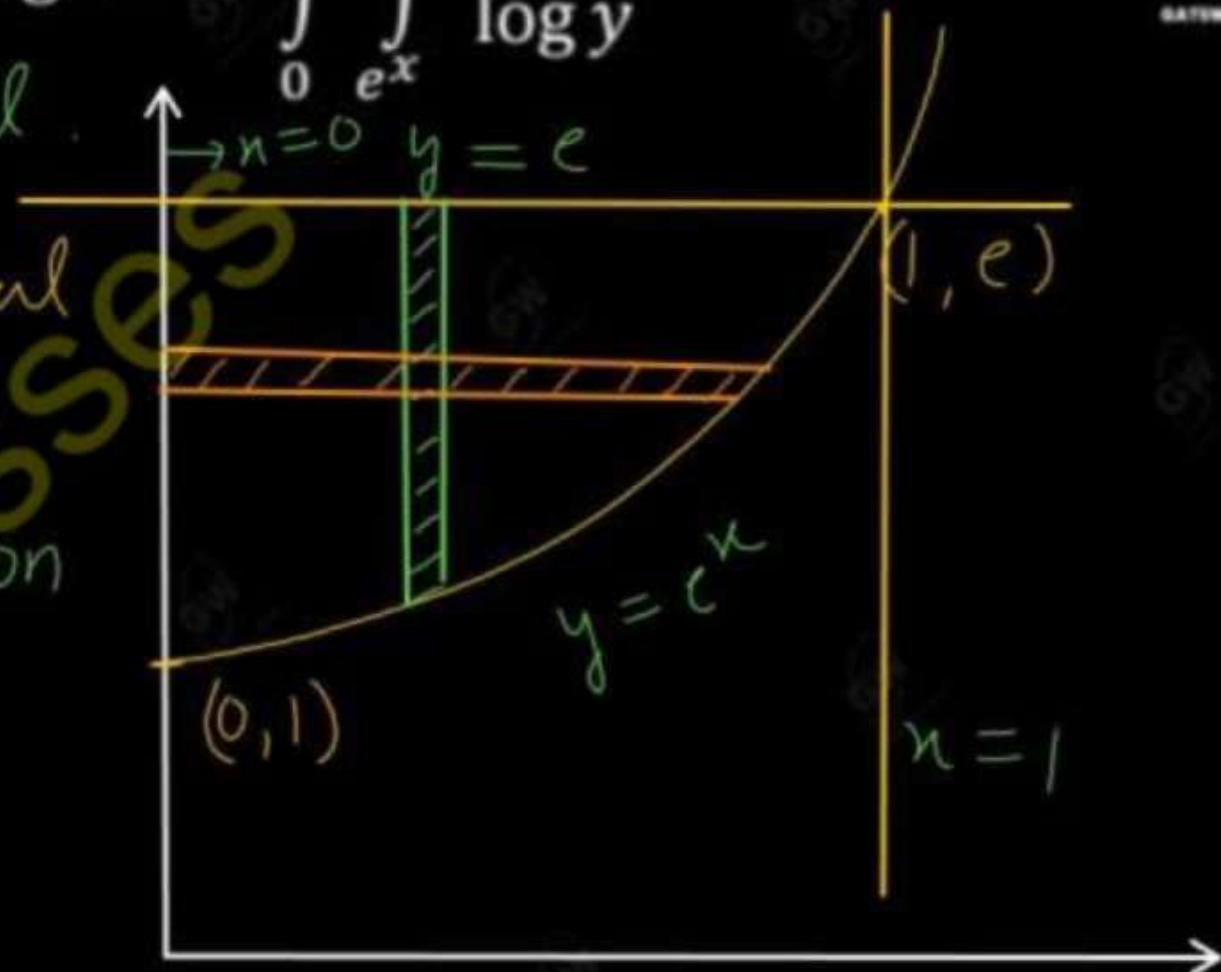
$$x = 0$$

$$e^x = y$$

$$\log e^x = \log y$$

$$x \log e = \log y \Rightarrow$$

$$x = \log y$$





Change the order of integration

$$\int_0^1 \int_{en}^e \frac{dy dx}{\log y} = \int_1^e \int_0^{\log y} \frac{dx dy}{\log y}$$

$$= \int_1^e \frac{dy}{\log y} (x)_0^{\log y}$$

$$= \int_1^e \frac{dy}{\log y} \log y$$

$$= \int_1^e dy$$

$$= (y)_1^e$$

$$\int_0^1 \int_{en}^e \frac{dy dx}{\log y} = e - 1$$

Q.3 Evaluate the following integral by changing the order of integration  $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$

AKTU 2021

$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$$

Given limits of integration

$$x = 0$$

$$x = \infty$$

$$y = x \checkmark$$

$$y = \infty$$

$\Rightarrow$  Given strip is vertical

Take a new Horizontal Strip

New limits of integration

$$y = 0$$

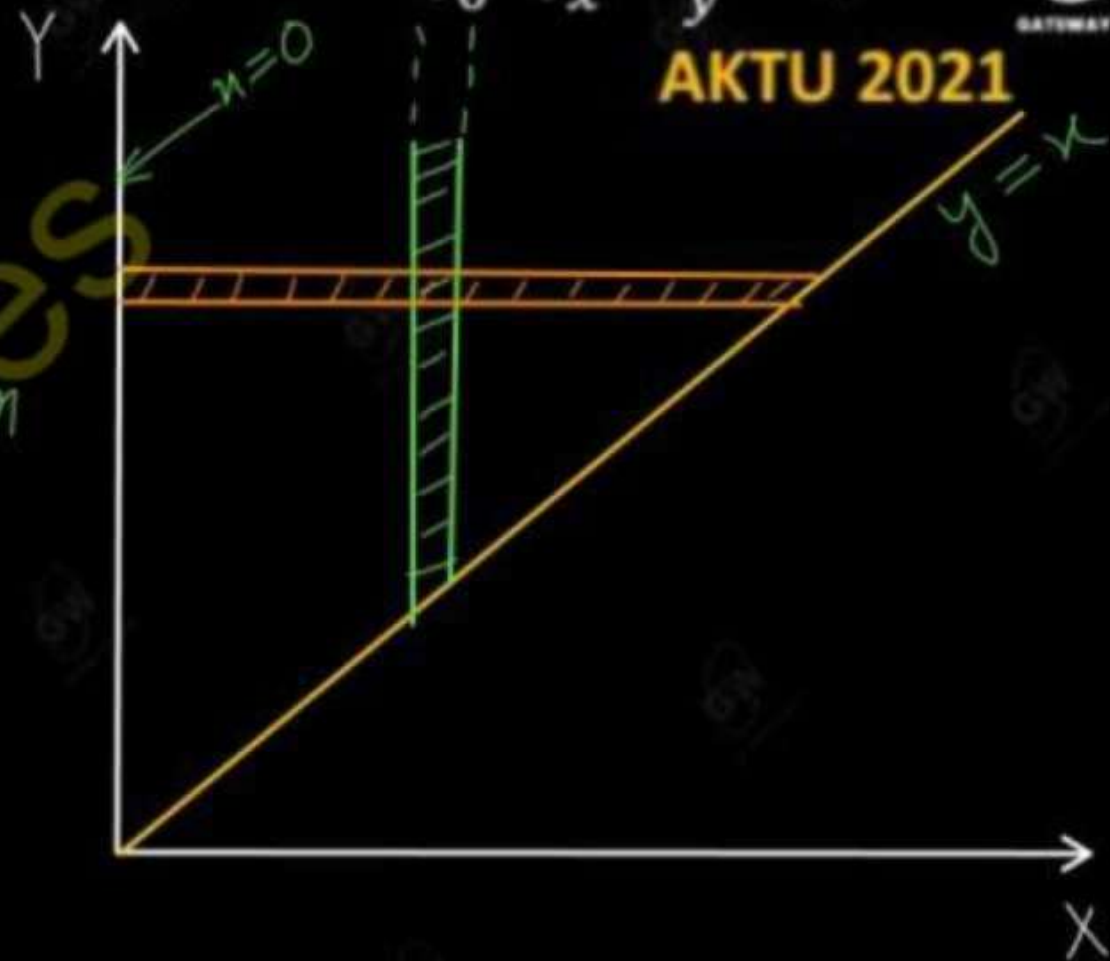
$$y = \infty$$

$$x = 0$$

$$x = y$$

Change the order of integration

$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx = \int_0^\infty \int_0^y \frac{e^{-y}}{y} dx dy$$





$$= \int_0^{\infty} \frac{e^{-y}}{y} dy \int_0^y dn$$

$$= \int_0^{\infty} \frac{e^{-y}}{y} dy (n)_0^y$$

$$= \int_0^{\infty} \frac{e^{-y}}{\cancel{y}} dy (\cancel{y})$$

$$= \int_0^{\infty} e^{-y} dy$$

$$= \left( \frac{e^{-y}}{-1} \right)_0^{\infty}$$

$$= -1 (e^{-y})_0^{\infty}$$

$$= -1 (e^{-\infty} - e^0)$$

$$= -1 (0 - 1)$$

$$= 1$$

$$\int_0^{\infty} \int_n^{\infty} \frac{e^{-y}}{y} dy dn = 1$$

Q.4 Evaluate the integrals by changing the order of integration  $\int_0^{\infty} \int_0^x x e^{-\frac{x^2}{y}} dy dx$

$$\int_0^{\infty} \int_0^x x e^{-\frac{x^2}{y}} dy dx$$

Take a new horizontal strip

Given limits of integration New limits of integration

$$x = 0$$

$$x = \infty$$

$$y = 0$$

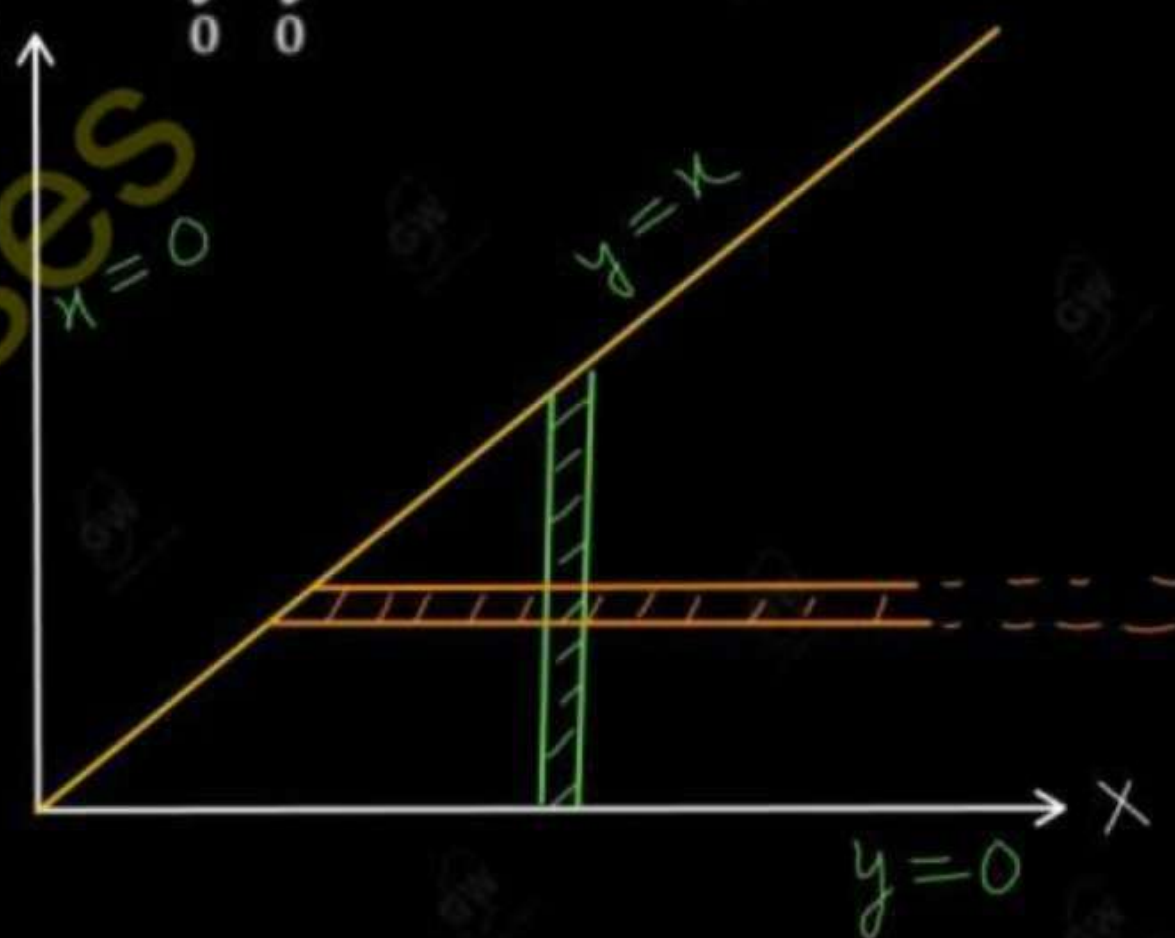
$$y = x \checkmark$$

$$y = 0$$

$$y = \infty$$

$$x = y$$

$$x = \infty$$



$\Rightarrow$  Given strip is vertical Change the order of integration



$$\int_0^{\infty} \int_0^n n e^{-\frac{n^2}{y}} dy dn = \int_0^{\infty} \int_y^{\infty} n e^{-\frac{n^2}{y}} dy dn$$

$$= \int_0^{\infty} dy \int_y^{\infty} n e^{-\frac{n^2}{y}} dn$$

Put  $\frac{n^2}{y} = t$

When  $n=y$ ,  $t=y$

When  $n=\infty$ ,  $t=\infty$

$$\frac{2n dn}{y} = dt$$

$$n dn = \frac{y}{2} dt$$

$$= \int_0^{\infty} dy \times \frac{y}{2} \int_y^{\infty} e^{-t} dt$$

$$= \int_0^{\infty} \frac{y}{2} dy \left( \frac{e^{-t}}{-1} \right)_y^{\infty}$$

$$= \frac{1}{2} \int_0^{\infty} y dy (-1) [e^{-\infty} - e^{-y}]$$

$$= \frac{1}{2} \int_0^{\infty} y dy (0 - e^{-y})$$

$$= \frac{1}{2} \int_0^{\infty} y e^{-y} dy$$

$$= \frac{1}{2} \left[ (y) (-e^{-y}) - (1) (e^{-y}) \right]_0^{\infty}$$

$$= \frac{1}{2} \left[ -y e^{-y} - e^{-y} \right]_0^{\infty}$$

$$= -\frac{1}{2} \left[ y e^{-y} + e^{-y} \right]$$

$$= -\frac{1}{2} \left[ 0 + 0 - (0 + 1) \right]$$

$$= \frac{1}{2}$$

$$\int_0^{\infty} \int_0^x n e^{-\frac{n^2}{y}} dy dn = \frac{1}{2}$$



UNIT -4 : DPP - 3

TOPIC : CHANGE OF ORDER OF INTEGRATION

Q.1 Evaluate the integrals by changing the order of integration  $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x \, dy \, dx}{\sqrt{x^2 + y^2}}$

Q.2 Evaluate the integrals by changing the order of integration  $\int_0^3 \int_0^{6/x} x^2 \, dy \, dx$ .

AKTU 2015

Q.3 Evaluate  $\int_0^2 \int_1^{e^x} dy \, dx$  by changing the order of integration

Q.4 Evaluate the integrals by changing the order of integration  $\int_0^1 \int_x^1 \sin y^2 \, dy \, dx$

AKTU 2015

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 4

#### Today's target

- TRIPLE INTEGRAL
- PYQs
- DPP



By Gulshan sir



$$\iiint_R f(x, y, z) dV = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{f_1(x, y)}^{f_2(x, y)} f(x, y, z) dz dy dx$$

$x \quad y \quad z$

Where R is the region bounded by the curves

$$a \leq x \leq b$$

$$\phi_1(x) \leq y \leq \phi_2(x)$$

$$f_1(x, y) \leq z \leq f_2(x, y)$$

NOTE :

- (1) Limits involving two variables kept innermost.
- (2) Limits involving one variables kept in middle.
- (3) Limits involving constants kept outermost
- (4) Start integration from innermost integral.
- (4) If all the limits are constant then the order of integration is not important.

Q. 1 Evaluate the following integral

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \underline{dx} \underline{dy} \underline{dz}$$

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \underline{dx} \underline{dy} \underline{dz}$$

$$= \int_0^1 \int_0^1 \int_0^1 e^x x e^y y e^z z \underline{dx} \underline{dy} \underline{dz}$$

$$= \int_0^1 \int_0^1 e^y x e^z dy dz \int_0^1 e^x dx$$

$$= \int_0^1 \int_0^1 e^y x e^z dy dz (e^x)'_0^1$$

$$= \int_0^1 \int_0^1 e^y x e^z dy dz (e-1)$$

$$= (e-1) \int_0^1 \int_0^1 e^y x e^z dy dz$$

$$= (e-1) \int_0^1 e^z dz \int_0^1 e^y dy$$

$$= (e-1) \int_0^1 e^z dz (e^y)'_0^1$$

$$= (e-1) \int_0^1 e^z dz (e-1)$$

$$= (e-1)^2 \int_0^1 e^z dz$$

$$= (e-1)^2 (e^z)'_0^1$$

$$= (e-1)^3$$



Q.2 Evaluate  $\int \int \int_R (x + y + z) dx dy dz$  where  $R : 0 \leq x \leq 1, 1 \leq y \leq 2, 2 \leq z \leq 3$

AKTU 2016, 2018

$$= \int_2^3 \int_1^2 \int_0^1 (x + y + z) dx dy dz$$

$$= \int_2^3 dz \int_1^2 \left( \frac{1}{2} + y + z \right) dy$$

$$= \left( 2z + \frac{y^2}{2} \right)_1^2$$

$$= \int_2^3 \int_1^2 dy dz \int_0^1 (x + y + z) dx$$

$$= \int_2^3 dz \left( \frac{1}{2} y + \frac{y^2}{2} + zy \right)_1^2$$

$$= 6 + \frac{9}{2} - 4 - 2$$

$$= \int_2^3 \int_1^2 dy dz \left( \frac{x^2}{2} + yx + zx \right)_0^1$$

$$= \int_2^3 dz \left[ 1 + 2 + 2z - \frac{1}{2} - \frac{1}{2} - z \right]$$

$$= \frac{9}{2}$$

$$= \int_2^3 \int_1^2 dy dz \left( \frac{1}{2} + y + z \right)$$

$$= \int_2^3 (2 + z) dz$$

Q.3 If the volume of an object expressed in the spherical coordinates as the following

$$V = \int_0^{2\pi} \int_0^{\pi} \int_0^1 r^2 \sin\phi \, dr \, d\phi \, d\theta$$

AKTU 2017

$$V = \int_0^{2\pi} \int_0^{\pi} \int_0^1 r^2 \sin\phi \, dr \, d\phi \, d\theta$$

$$V = \int_0^{2\pi} \int_0^{\pi} \sin\phi \, d\phi \, d\theta \int_0^1 r^2 \, dr$$

$$V = \int_0^{2\pi} \int_0^{\pi} \sin\phi \, d\phi \, d\theta \left( \frac{r^3}{3} \right)_0^1$$

$$V = \int_0^{2\pi} \int_0^{\pi} \sin\phi \, d\phi \, d\theta \left( \frac{1}{3} - 0 \right)$$

$$V = \frac{1}{3} \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi \, d\phi$$

$$V = \frac{1}{3} \int_0^{2\pi} d\theta \left( -\cos\phi \right)_0^{\pi}$$

$$V = -\frac{1}{3} \int_0^{2\pi} d\theta \left( \cos\phi \right)_0^{\pi}$$

$$V = -\frac{1}{3} \int_0^{2\pi} d\theta (-1 - 1)$$

$$V = \frac{2}{3} \int_0^{2\pi} d\theta$$

$$V = \frac{2}{3} \left( \theta \right)_0^{2\pi}$$

$$V = \frac{2}{3} (2\pi - 0)$$

$$V = \frac{4\pi}{3}$$



Q. 4 Find the value of

$$\int_0^1 \int_0^x \int_0^{x+y} dx dy dz$$

$$\int_0^1 \int_0^x \int_0^{x+y} dz dy dx$$

$$= \int_0^1 \int_0^x dz dy dx$$

$$= \int_0^1 \int_0^x (z)_0^{x+y} dy dx$$

$$= \int_0^1 \int_0^x (x+y) dy dx$$

$$= \int_0^1 dx \int_0^x (x+y) dy$$

$$= \int_0^1 dx \left( xy + \frac{y^2}{2} \right)_0^x$$

$$= \int_0^1 dx \left( x^2 + \frac{x^2}{2} - 0 \right)$$

$$= \int_0^1 \frac{3}{2} x^2 dx$$

AKTU 2022

$$= \frac{3}{2} \int_0^1 x^2 dx$$

$$= \frac{3}{2} \left( \frac{x^3}{3} \right)_0^1$$

$$= \frac{3}{2} \times \frac{1}{3}$$

$$= \frac{1}{2}$$

Q.5 Evaluate  $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} dz dy dx$ .

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} dz dy dx$$

$x \quad y \quad z$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} dy dx \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{\left(\sqrt{1-x^2-y^2}\right)^2 - z^2}} dz$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} dy dx \left( \sin^{-1} \frac{z}{\sqrt{1-x^2-y^2}} \right)_0^{\sqrt{1-x^2-y^2}}$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} dy dx (\sin^{-1} 1 - \sin^{-1} 0)$$

$$= \frac{\pi}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} dy dx$$

$$= \frac{\pi}{2} \int_0^1 dx \left( y \right)_0^{\sqrt{1-x^2}}$$

$$= \frac{\pi}{2} \int_0^1 \sqrt{1-x^2} dx$$

$$= \frac{\pi}{2} \left[ \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right]_0^1$$

$$= \frac{\pi}{2} \left[ 0 + \frac{1}{2} \sin^{-1} 1 - 0 - 0 \right]$$

$$= \frac{\pi}{2} \left( \frac{\pi}{4} \right)$$

$$= \frac{\pi^2}{8}$$



Q. 6 Evaluate

$$\int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dy dx dz.$$

$$\int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dy dx dz$$

$z \quad x \quad y$

$$= \int_0^4 \int_0^{2\sqrt{z}} dx dz \left( y \right)_0^{\sqrt{4z-x^2}}$$

$$= \int_0^4 \int_0^{2\sqrt{z}} dx dz (\sqrt{4z-x^2} - 0)$$

$$= \int_0^4 \int_0^{2\sqrt{z}} \sqrt{4z-x^2} dx dz$$

$$= \int_0^4 dz \int_0^{2\sqrt{z}} \sqrt{(2\sqrt{z})^2 - x^2} dx$$

$$= \int_0^4 dz \left[ \frac{x}{2} \sqrt{4z-x^2} + \frac{4z}{2} \sin^{-1} \frac{x}{2\sqrt{z}} \right]_0^{2\sqrt{z}}$$

$$= \int_0^4 dz (0 + 2z \sin^{-1} 1 - 0)$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$= \frac{\pi}{2} \int 2z dz$$

$$= \pi \int_0^4 z dz$$

$$= \pi \left( \frac{z^2}{2} \right)_0^4$$

$$= \pi \times \frac{16}{2}$$

$$= 8\pi$$

UNIT -4 : DPP - 4

TOPIC : TRIPLE INTEGRAL

**Q.1** Evaluate  $\int \int \int_R (x - 2y + z) dz dy dx$  where R is the region determined by  $0 \leq x \leq 1$ ,

$$0 \leq y \leq x^2, 0 \leq z \leq x + y$$

AKTU 2023

**Q.2** Evaluate the following integral  $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz dy dx$

AKTU 2017

**Q.3** Evaluate the following integral  $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$ .



## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 5

#### Today's target

- Double integral using change of variables ( $u - v$  form)
- PYQs
- DPP



By Gulshan sir

- Some times the evaluation of double integral or triple integral become easy after suitable change of variables

### Double integral using change of variables ( $u - v$ form)

Let  $\int \int_R f(x, y) \underline{dx dy}$

where  $R$  is the region bounded by the lines

$$\begin{cases} a_1x + a_2y = c_1 \\ a_1x + a_2y = c_2 \end{cases}$$

$$\begin{cases} b_1x + b_2y = d_1 \\ b_1x + b_2y = d_2 \end{cases}$$

Let  $a_1x + a_2y = u$  ————— (1)

$a_1x + a_2y = v$  ————— (2)

Solve (1) and (2)

$$x = \phi_1(u, v)$$

$$y = \phi_2(u, v)$$

$$J = \frac{\partial(x, y)}{\partial(u, v)}$$

$$dx dy = |J| du dv$$

Limits of integration

$$u = c_1 \text{ to } u = c_2$$

$$v = d_1 \text{ to } v = d_2$$

$$\int \int_R f(x, y) dx dy = \int_{d_1}^{d_2} \int_{c_1}^{c_2} f(u, v) |J| du dv$$



Q. 1 Evaluate  $\iint_D (y - x) dx dy$  where D is the region in xy - plane bounded by the straight lines  $y = x + 1$ ,  $y = x - 3$ ,  $y = -\frac{1}{3}x + \frac{7}{3}$ ,  $y = -\frac{1}{3}x + 5$

AKTU 2024

$$\iint_D (y - x) dx dy$$

D is the region bounded by

$$y = x + 1 \Rightarrow y - x = 1$$

$$y = x - 3 \Rightarrow y - x = -3$$

$$y = -\frac{1}{3}x + \frac{7}{3} \Rightarrow y + \frac{1}{3}x = \frac{7}{3}$$

$$y = -\frac{1}{3}x + 5 \Rightarrow y + \frac{1}{3}x = 5$$

Let

$$y - x = u \quad \text{--- (1)}$$

$$y + \frac{1}{3}x = v \quad \text{--- (2)}$$

Solve (1) and (2)

$$y - x - y - \frac{1}{3}x = u - v$$

$$-x - \frac{1}{3}x = u - v$$

$$-\frac{4}{3}x = u - v$$

$$-x = \frac{3}{4}u - \frac{3}{4}v$$

$$x = \frac{3}{4}v - \frac{3}{4}u$$

Put x in (1)

$$y = x + u$$

$$y = \frac{3}{4}v - \frac{3}{4}u + u$$

$$y = \frac{3}{4}v + \frac{1}{4}u$$

$$J = \frac{\partial(x, y)}{\partial(u, v)}$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J = \begin{vmatrix} -\frac{3}{4} & \frac{3}{4} \\ \frac{1}{4} & \frac{3}{4} \end{vmatrix}$$

$$J = -\frac{3}{4} \times \frac{3}{4} - \frac{1}{4} \times \frac{3}{4}$$

$$J = -\frac{9}{16} - \frac{3}{16}$$

$$J = -\frac{12}{16} = -\frac{3}{4}$$

$$J = -\frac{3}{4}$$

$$|J| = \frac{3}{4}$$

$$dx dy = |J| du dv$$

Limits of integration

$$u = -3 \text{ to } u = 1$$

$$v = \frac{7}{3} \text{ to } v = 5$$

$$\iint_D (y-x) dx dy = \int_{\frac{7}{3}}^5 \int_{-3}^1 u |J| du dv$$

$$= \int_{\frac{7}{3}}^5 \int_{-3}^1 u \times \frac{3}{4} du dv$$

$$= \frac{3}{4} \int_{\frac{7}{3}}^5 dv \int_{-3}^1 u du$$

$$= \frac{3}{4} \int_{\frac{7}{3}}^5 dv \left( \frac{u^2}{2} \right)_{-3}^1$$



$$\int \int_D (y-x) dx dy = \frac{3}{4} \int_{7/3}^5 dv \left( \frac{1}{2} - \frac{9}{2} \right)$$

$$= \frac{3}{4} \times (-4) \int_{7/3}^5 dv$$

$$= -3 \left( v \right)_{7/3}^5 = -3 \left( 5 - \frac{7}{3} \right)$$

$$= -3 \times \frac{8}{3}$$

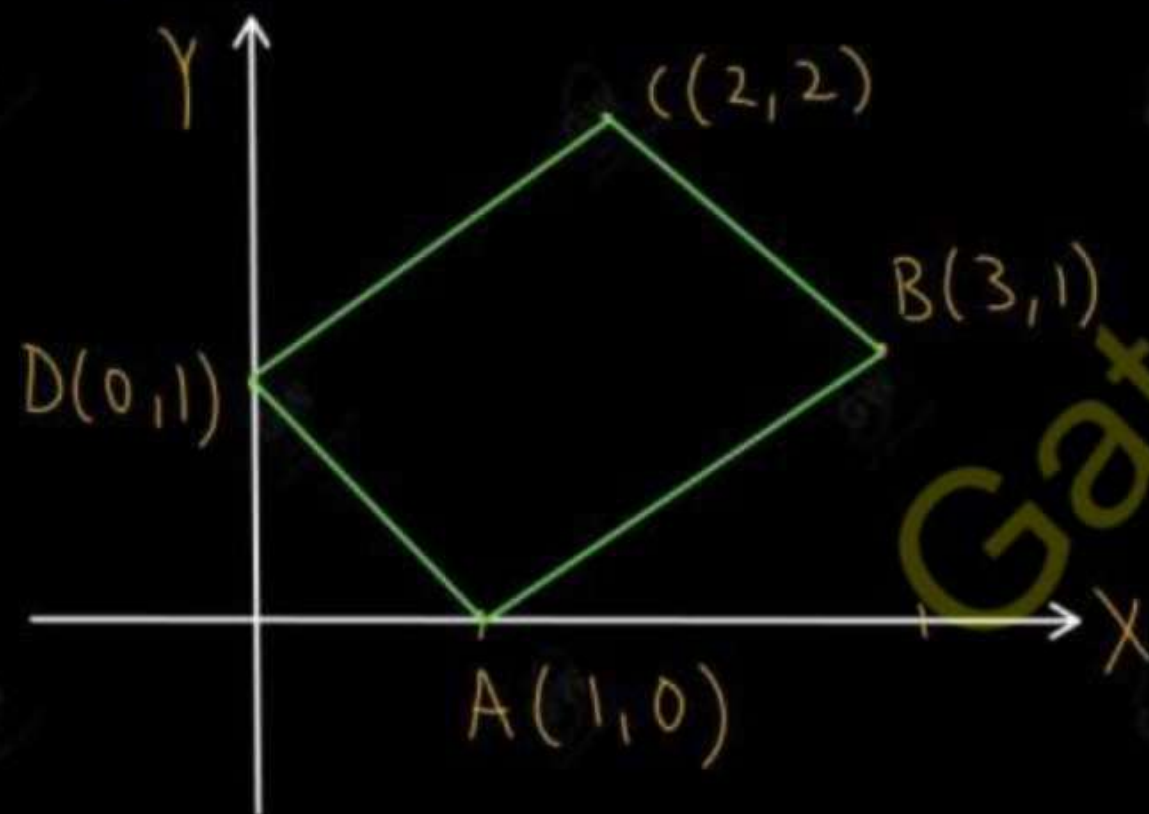
$$= -8$$

**Q. 2** Evaluate  $\iint_R (x+y)^2 dx dy$  where  $R$  is the parallelogram in the  $xy$  - plane with vertices  $(1, 0), (3, 1), (2, 2), (0, 1)$  using the transformation  $u = x + y$  and  $v = x - 2y$

$$\iint_R (x+y)^2 dx dy$$

$R$  is a  $\parallel^m ABCD$

$A(1, 0) B(3, 1) C(2, 2) D(0, 1)$



New region of integration

$$u = x + y \quad \text{--- (1)}$$

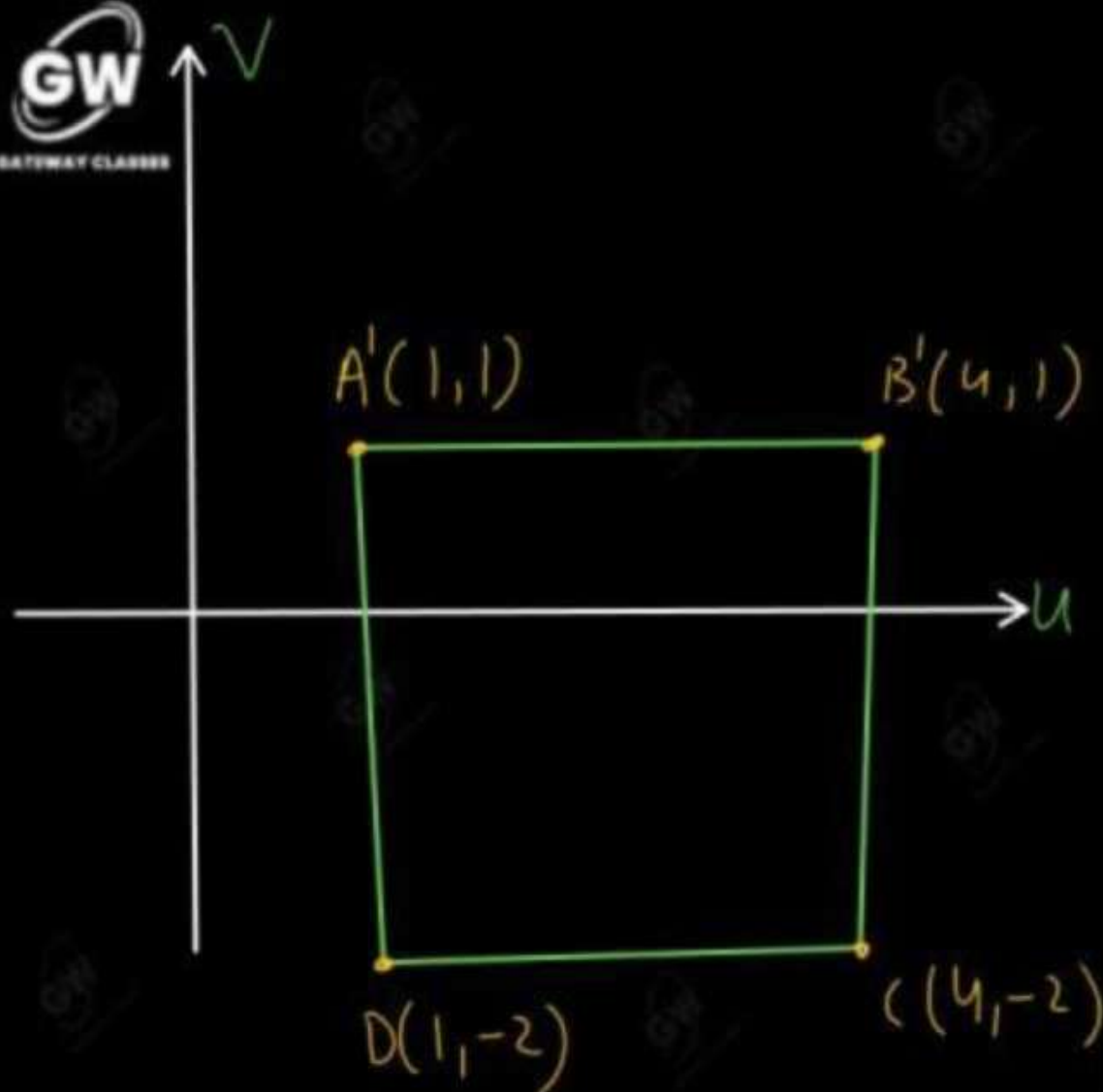
$$v = x - 2y \quad \text{--- (2)}$$

New vertices in  $u-v$  plane

| $xy$ plane | $A(1, 0)$  | $B(3, 1)$  | $C(2, 2)$   | $D(0, 1)$   |
|------------|------------|------------|-------------|-------------|
| $uv$ plane | $A'(1, 1)$ | $B'(4, 1)$ | $C'(4, -2)$ | $D'(1, -2)$ |

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Limits of integration

$$u = 1 \text{ to } u = 4$$

$$v = -2 \text{ to } v = 1$$

Solve ① and ②

$$u - v = \cancel{x} + y - \cancel{x} + 2y$$

$$3y = u - v$$

$$y = \frac{1}{3}u - \frac{1}{3}v$$

Put  $y$  in ①

$$u = x + y$$

$$x = u - y$$

$$x = u - \frac{1}{3}u + \frac{1}{3}v$$

$$x = \frac{2}{3}u + \frac{1}{3}v$$

$$J = \frac{\partial(x, y)}{\partial(u, v)}$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$J = \begin{vmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{vmatrix}$$

$$J = \frac{2}{3} \times \left(-\frac{1}{3}\right) - \frac{1}{3} \times \frac{1}{3}$$

$$J = -\frac{2}{9} - \frac{1}{9}$$

$$J = -\frac{1}{3}$$

$$|J| = \frac{1}{3}$$

$$\iint_R (x+y)^2 dx dy = \int_{-2}^1 \int_1^4 u^2 |J| du dv$$

$$= \int_{-2}^1 \int_1^4 u^2 \times \frac{1}{3} du dv$$

$$= \frac{1}{3} \int_{-2}^1 dv \int_1^4 u^2 du$$

$$\iint_R (x+y)^2 dx dy = \frac{1}{3} \int_{-2}^1 dv \left( \frac{u^3}{3} \right)_1^4$$

$$= \frac{1}{3} \int_{-2}^1 dv \left( \frac{64}{3} - \frac{1}{3} \right)$$

$$= \frac{1}{3} \times \frac{63}{3} \int_{-2}^1 dv = 7(v)_{-2}^1$$

$$= 7(1+2)$$

$$= 21$$



UNIT -4 : DPP - 5

**TOPIC :** Double integral using change of variables ( $u - v$  form)

**Q. 1** Evaluate by changing the variables,  $\iint_R (x + y)^2 dx dy$  where  $R$  is the region bounded by the parallelogram  $x + y = 0$ ,  $x + y = 2$ ,  $3x - 2y = 0$  and  $3x - 2y = 3$  **AKTU 2021**

**Q. 2** Evaluate  $\iint_R (x - y)^4 e^{x+y} dx dy$  where  $R$  is the square with vertices at  $(1, 0)$ ,  $(2, 1)$ ,  $(1, 2)$  and  $(0, 1)$

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 6

#### Today's target

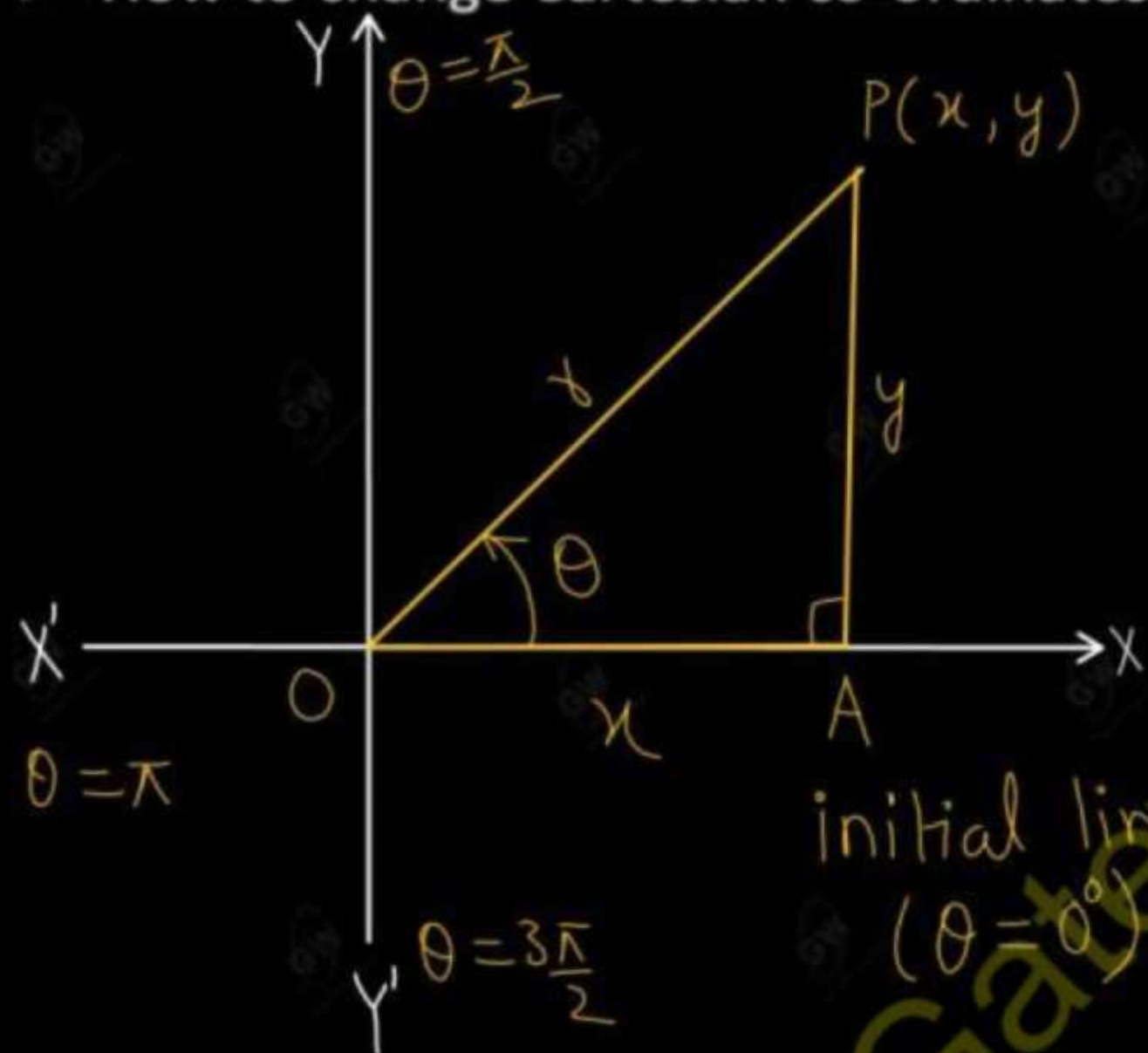
- Double integral using change of variables  
(Cartesian form to polar form)
- PYQs
- DPP



By Gulshan sir



➤ How to change Cartesian co-ordinates  $(x, y)$  in to polar coordinates  $(r, \theta)$



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$J = \frac{\partial(x, y)}{\partial(r, \theta)}$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$J = r \cos^2 \theta + r \sin^2 \theta$$

$$J = r$$

$$dx dy = |J| dr d\theta$$

$$dx dy = r dr d\theta$$

## STEP TO SOLVE QUESTIONS

**STEP – 1** : Draw the region for the given limits of integration

**STEP – 2** : Put  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $x^2 + y^2 = r^2$ ,  $dx dy = r dr d\theta$  in the given integration

**STEP – 3** : Find the limits of  $r$  and  $\theta$  from the region

**STEP – 4** : Now solve the polar form



Q. 1 Evaluate the following by changing into polar coordinates:  $\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} dx dy$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} dx dy$$

Given limits of integration

✓  $y = 0$  to  $y = a$

$x = 0$  to  $x = \sqrt{a^2-y^2}$

$$x^2 = a^2 - y^2$$

$$x^2 + y^2 = a^2$$

$\Rightarrow x = 0$  to  $x^2 + y^2 = a^2$

To change in polar coordinates

Put  $x = r \cos \theta$

$$y = r \sin \theta$$

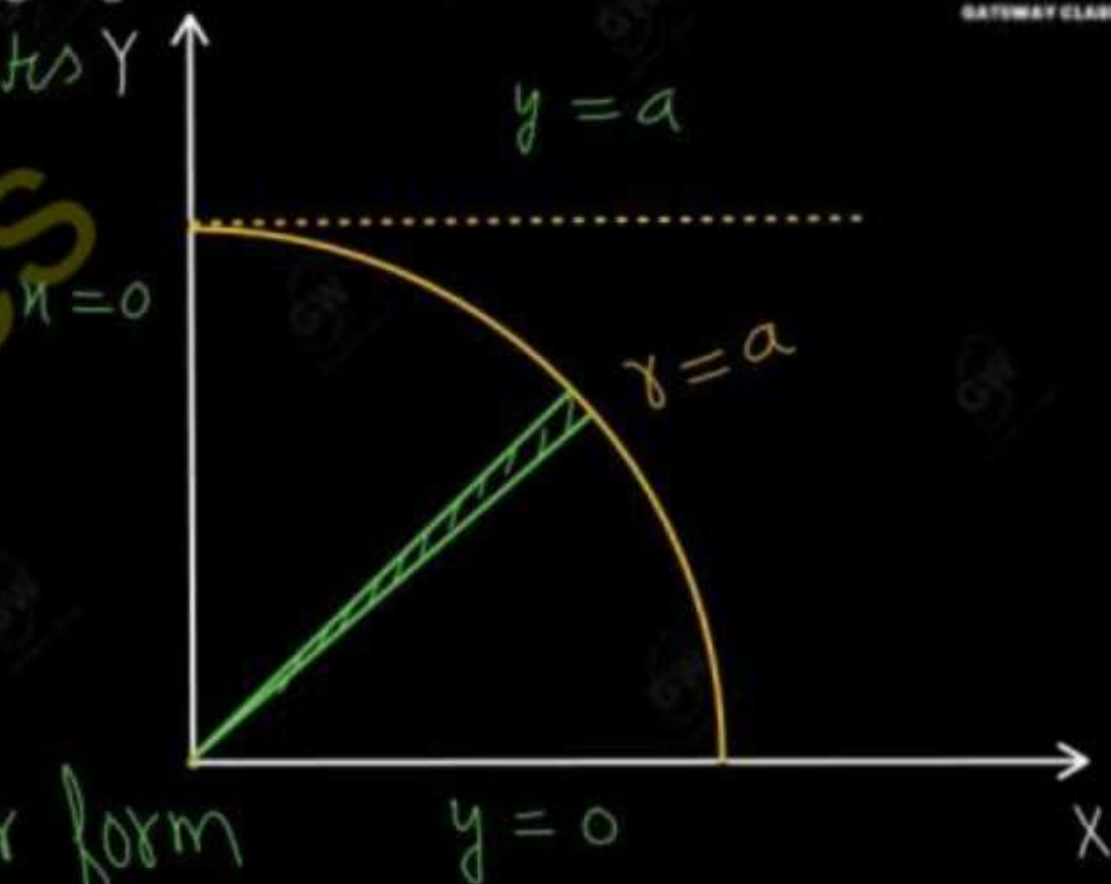
$$x^2 + y^2 = r^2$$

$$dx dy = r dr d\theta$$

Equation of circle in polar form

$$r^2 = a^2$$

$$r = a$$



Limits of integration

$$\theta = 0^\circ \text{ to } \theta = \frac{\pi}{2}$$

$$r = 0 \text{ to } r = a$$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} dx dy = \int_{\theta=0}^{\theta=\pi/2} \int_{r=0}^r (r \sin \theta)^2 \sqrt{r^2} r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a r^2 \sin^2 \theta \times r \times r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a r^4 \sin^2 \theta dr d\theta$$

$$= \int_0^{\pi/2} \sin^2 \theta d\theta \int_0^a r^4 dr$$

$$= \int_0^{\pi/2} \sin^2 \theta d\theta \left( \frac{r^5}{5} \right)_0^a$$

$$= \frac{a^5}{5} \int_0^{\pi/2} \sin^2 \theta d\theta$$

$$= \frac{a^5}{5} \int_0^{\pi/2} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \frac{a^5}{10} \int_0^{\pi/2} (1 - \cos 2\theta) d\theta$$



$$= \frac{a^5}{10} \left( \theta - \frac{\sin 2\theta}{2} \right)_0^{\pi/2}$$

$$= \frac{a^5}{10} \left( \frac{\pi}{2} - 0 - 0 - 0 \right)$$

$$= \frac{\pi a^5}{20}$$

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} \, dx \, dy = \frac{\pi a^5}{20}$$

Q. 2 Evaluate

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{xdydx}{\sqrt{x^2+y^2}} \text{ by changing into polar coordinates}$$

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{xdydx}{\sqrt{x^2+y^2}}$$

Given limits of integration

$$x=0 \text{ to } x=2$$

$$y=0 \text{ to } y=\sqrt{2x-x^2}$$

$$y^2 = 2x - x^2$$

$$x^2 + y^2 = 2x$$

$$x^2 - 2x + y^2 = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

$$(x-1)^2 + (y-0)^2 = 1^2$$

$$\Rightarrow y=0 \text{ to } (x-1)^2 + (y-0)^2 = 1^2$$

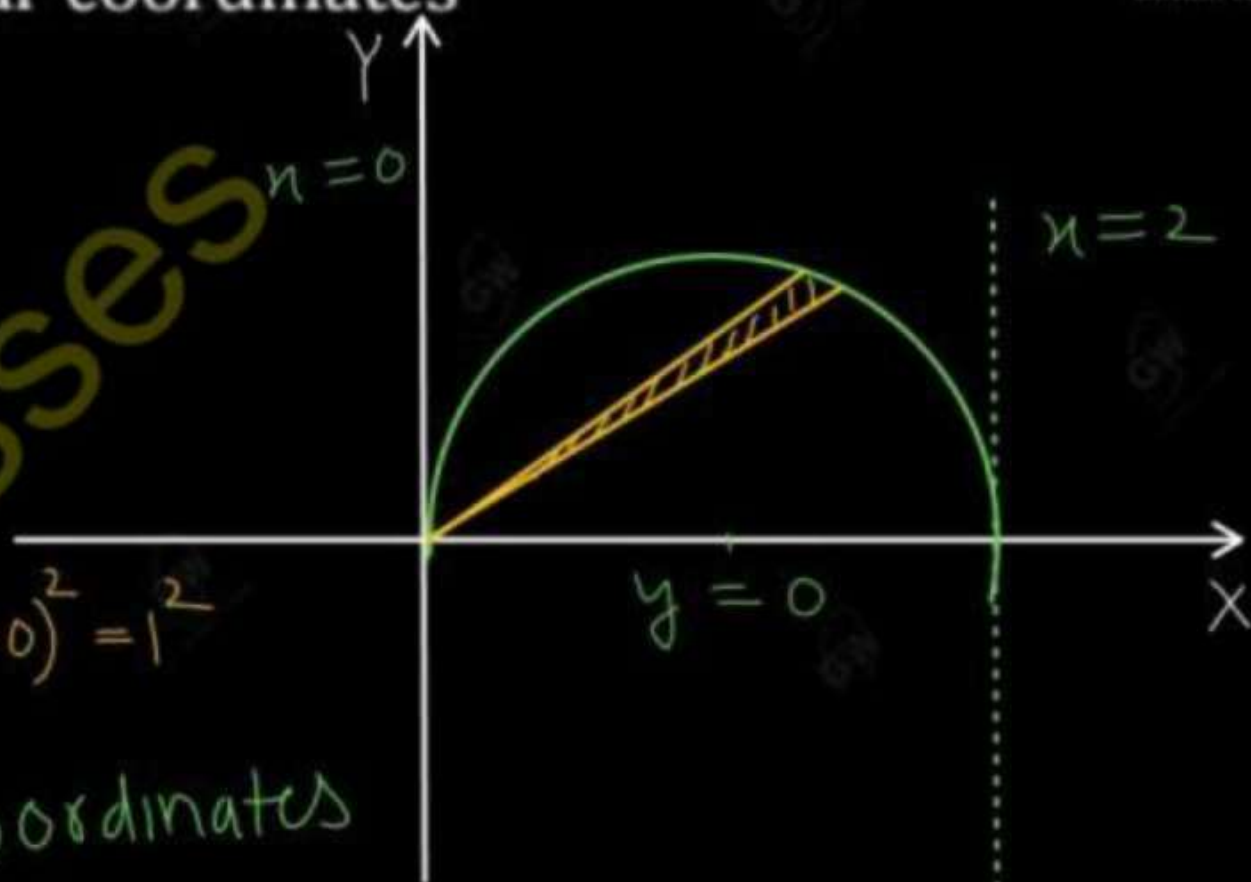
To change in polar coordinates

$$\text{Put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$dx dy = r dr d\theta$$



Equation of circle in polar form

$$x^2 + y^2 = 2x$$

$$(r \cos \theta)^2 + (r \sin \theta)^2 = 2r \cos \theta$$

$$r^2 = 2r \cos \theta$$



$$r = 2 \cos \theta$$

Limits of integration

$$\theta = 0 \text{ to } \theta = \frac{\pi}{2}$$

$$r = 0 \text{ to } r = 2 \cos \theta$$

$$\int_0^2 \int_0^{\sqrt{2n-n^2}} \frac{ndydn}{\sqrt{n^2+y^2}}$$

$$= \int_0^{\pi/2} \int_0^{2 \cos \theta} \frac{r \cos \theta \times r dr d\theta}{\sqrt{r^2}}$$

$$= \int_0^{\pi/2} \int_0^{2 \cos \theta} r \cos \theta dr d\theta$$

$$= \int_0^{\pi/2} \cos \theta d\theta \int_0^{2 \cos \theta} r dr$$

$$= \int_0^{\pi/2} \cos \theta d\theta \left( \frac{r^2}{2} \right)_0^{2 \cos \theta}$$

$$= \int_0^{\pi/2} \cos \theta d\theta \times \frac{4 \cos^2 \theta}{2}$$

$$= 2 \int_0^{\pi/2} \cos^3 \theta d\theta$$

$$= \frac{2}{4} \int_0^{\pi/2} (\cos 3\theta + 3 \cos \theta) d\theta$$

$$= \frac{1}{2} \left( \frac{\sin 3\theta}{3} + 3 \sin \theta \right)_0^{\pi/2}$$

$$= \frac{1}{2} \left( -\frac{1}{3} + 3 \right) = \frac{1}{2} \left( \frac{-1+9}{3} \right) = \frac{4}{3}$$

$$\int_0^2 \int_0^{\sqrt{2n-n^2}} \frac{ndydn}{\sqrt{n^2+y^2}} = \frac{4}{3}$$

Q. 3 Change into polar coordinates and evaluate  $\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx$ . Hence show

that  $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx$$

Given limits of integration

$$x = 0 \text{ to } x = \infty$$

$$y = 0 \text{ to } y = \infty$$

To change in polar form

Put  $x = r \cos \theta$

$$y = r \sin \theta$$

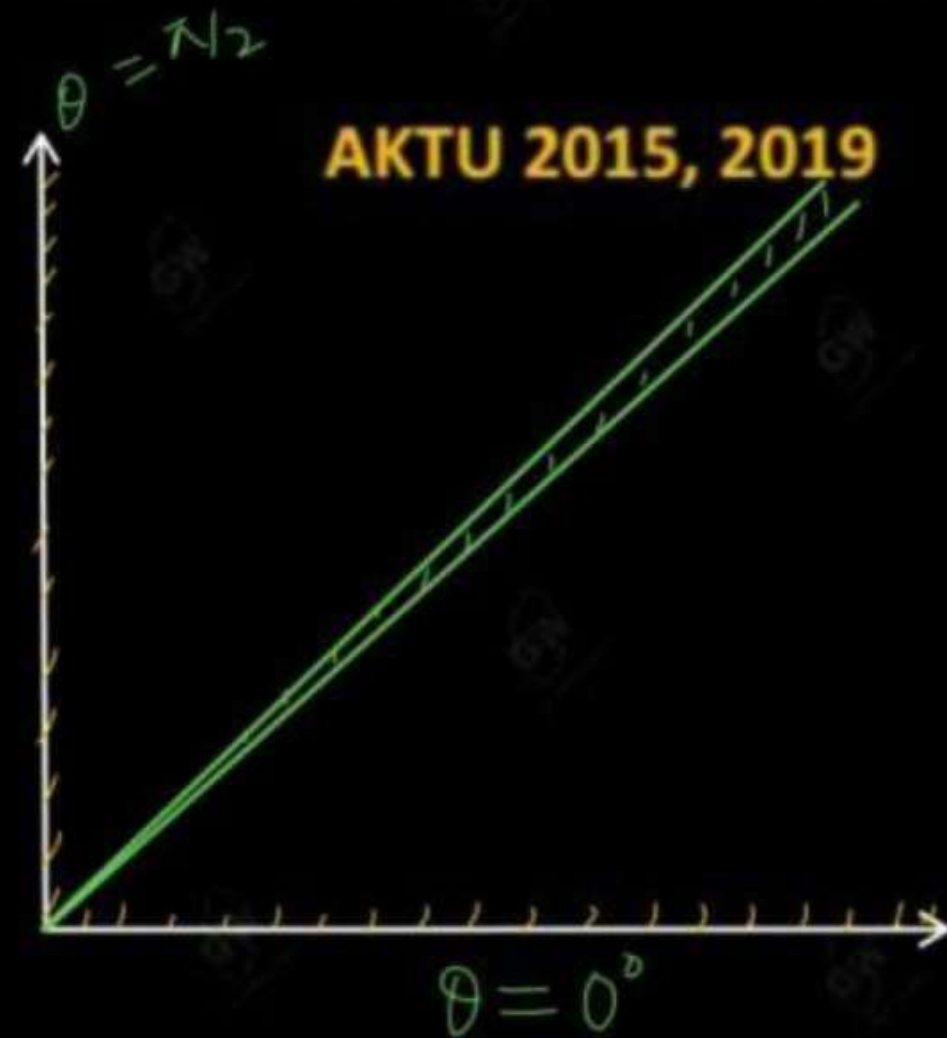
$$x^2 + y^2 = r^2$$

$$dx dy = r dr d\theta$$

New limits of integration

$$\theta = 0 \text{ to } \theta = \pi/2$$

$$r = 0 \text{ to } r = \infty$$





$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx$$

$$= \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r dr d\theta$$

Put  $r^2 = t$

$$2r dr = dt$$

$$r dr = \frac{dt}{2}$$

When  $r=0$ ,  $t=0$

$r=\infty$ ,  $t=\infty$

$$= \int_0^{\pi/2} \int_0^{\infty} e^{-t} \frac{dt}{2} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta \int_0^{\infty} e^{-t} dt$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta \left( -e^{-t} \right)_0^{\infty}$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta \left( e^{-\infty} - e^0 \right)$$

$$= -\frac{1}{2} (0-1) \int_0^{\pi/2} d\theta = \frac{1}{2} (\theta)_0^{\pi/2}$$

$$= \frac{\pi}{4}$$

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx = \frac{\pi}{4}$$

Let  $I = \int_0^{\infty} e^{-x^2} dx$  ——— (1)

$I = \int_0^{\infty} e^{-y^2} dy$  ——— (2)

Multiply ① and ②

$$I \times I = \int_0^{\infty} e^{-x^2} dx \int_0^{\infty} e^{-y^2} dy$$

$$I^2 = \int_0^{\infty} \int_0^{\infty} e^{-x^2} \times e^{-y^2} dx dy$$

$$I^2 = \int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy$$

$$I^2 = \frac{\pi}{4}$$

$$I = \frac{\sqrt{\pi}}{2}$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Hence proved



**Q. 4** Evaluate the integral  $\iint_R \sqrt{x^2 + y^2} \, dx \, dy$  by changing to polar coordinates where  $R$  is the region in the  $xy$  - plane bounded by the circles  $x^2 + y^2 = 4$  and  $x^2 + y^2 = 9$ .

$$\iint_R \sqrt{x^2 + y^2} \, dx \, dy$$

Where  $R$  is bounded by

$$x^2 + y^2 = 4$$

$$x^2 + y^2 = 9$$

To change in polar form

$$\text{Put } x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$dx \, dy = r \, dr \, d\theta$$

Equation of circle  
in polar form

$$r^2 = 4$$

$$r = 2$$

$$r^2 = 9$$

$$r = 3$$



New limits of integration

$$\theta = 0 \text{ to } \theta = 2\pi$$

$$r = 2 \text{ to } r = 3$$

$$\iint_R \sqrt{x^2 + y^2} \, dx \, dy$$

$$= \int_0^{2\pi} \int_2^3 \sqrt{r^2} \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_2^3 r^2 \, dr \, d\theta$$

$$= \int_0^{2\pi} d\theta \int_2^3 r^2 \, dr$$

$$= \int_0^{2\pi} d\theta \left( \frac{r^3}{3} \right)_2^3$$

$$= \int_0^{2\pi} d\theta \left( \frac{27}{3} - \frac{8}{3} \right)$$

$$= \frac{19}{3} (\theta)_0^{2\pi}$$

$$= \frac{19}{3} (2\pi - 0)$$

$$= \frac{38}{3} \pi$$

$$\boxed{\iint_R \sqrt{x^2 + y^2} \, dx \, dy = \frac{38}{3} \pi}$$



UNIT -4 : DPP - 6

**TOPIC :** Double integral using change of variables (Cartesian form to polar form)

**Q.1** Evaluate  $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2 + y^2) dx dy$  by changing into polar coordinates

**Q.2** Evaluate  $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2 + y^2) dx dy$  by changing into polar coordinates

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 7

#### Today's target

- BETA AND GAMMA FUNCTION (PART – 1)
- PYQs
- DPP



By Gulshan sir



## Gamma Function ( $\Gamma$ )

The Gamma Function of  $n$  is defined as

$$\int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma n$$

$$n > 0$$

Conclusion :

$$\int_0^{\infty} e^{-x} x^n dx = \Gamma n + 1$$

Important Results

$$(i) \Gamma 1 = 1$$

$$(ii) \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$(iii) \Gamma 0 = \infty$$

$$\int_0^{\infty} e^{-kx} x^{n-1} dx = \frac{\Gamma n}{k^n}$$

Proof :

We know that

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx$$

Put  $x = ky$   
 $dx = k dy$

When  $x = 0$ ,  $y = 0$

When  $x = \infty$ ,  $y = \infty$

$$\Gamma n = \int_0^{\infty} e^{-ky} (ky)^{n-1} k dy$$

$$\Gamma n = \int_0^{\infty} e^{-ky} k^{n-1} y^{n-1} k dy$$

$$\Gamma n = k^n \int_0^{\infty} e^{-ky} y^{n-1} dy$$

$$\int_0^{\infty} e^{-ky} y^{n-1} dy = \frac{\Gamma n}{k^n}$$

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Using

$$\int_a^b f(x) dx = \int_a^b f(t) dt$$

$$\Rightarrow \int_0^{\infty} e^{-kx} x^{n-1} dx = \frac{\Gamma n}{k^n}$$

Henu proved



$$\Gamma n + 1 = n \Gamma n$$

Proof :

We know that

$$\Gamma n + 1 = \int_0^{\infty} \underset{\text{II}}{e^{-x}} \underset{\text{I}}{x^n} dx$$

$$\Gamma n + 1 = x^n \int_0^{\infty} e^{-x} dx - \int_0^{\infty} n x^{n-1} (-e^{-x}) dx$$

$$\Gamma n + 1 = \left( -x^n e^{-x} \right)_0^{\infty} + n \int_0^{\infty} e^{-x} x^{n-1} dx$$

$$\Gamma n + 1 = (0 - 0) + n \Gamma n$$

$$\Gamma n + 1 = n \Gamma n$$

Hence proved

Conclusion

$$\Gamma n = (n-1) \Gamma n-1$$

If  $n$  is a positive integer, then repeated application of reduction formula we get

$$\Gamma n + 1 = n!$$

Proof: We know that

$$\Gamma n + 1 = n \Gamma n$$

$$= n(n-1) \Gamma n-1$$

$$= n(n-1)(n-2) \Gamma n-2$$

$$= n(n-1)(n-2)(n-3) \Gamma n-3$$

$$= n(n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1$$

$$= n!$$

$$\Gamma n + 1 = n!$$

conclusion

$$\Gamma n = (n-1)!$$



(1) If  $n$  is a positive integer

$$\Gamma n = (n - 1)!$$

(2) If  $n$  is a positive fraction

$$\Gamma n = (n - 1)\Gamma n - 1$$

(3) If  $n$  is a negative fraction

$$\Gamma n = \frac{\Gamma n+1}{n}$$

(4) If  $n$  is a negative integer

$$\Gamma n = \frac{\Gamma n+1}{n}$$

NOTE :  $\Gamma n = \infty$

Q.1 Evaluate the following

(i)  $\Gamma 6$

(ii)  $\Gamma \frac{3}{2}$

(iii)  $\Gamma \frac{5}{2}$

(iv)  $\Gamma \frac{7}{2}$

(v)  $\Gamma \frac{9}{2}$

(vi)  $\Gamma \frac{-1}{2}$

(vii)  $\Gamma \frac{-5}{2}$

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(viii)  $\Gamma - 3$

$$\begin{aligned} (i) \Gamma 6 &= (6-1)! \\ &= 5! \\ &= 5 \times 4 \times 3 \times 2 \times 1 \\ &= 120 \end{aligned}$$

$$\begin{aligned} (ii) \Gamma \frac{3}{2} &= \left(\frac{3}{2}-1\right) \Gamma \frac{3}{2}-1 \\ &= \frac{1}{2} \Gamma \frac{1}{2} \\ &= \frac{1}{2} \sqrt{\pi} \end{aligned}$$

$$\begin{aligned} (iii) \Gamma \frac{5}{2} &= \frac{3}{2} \times \frac{1}{2} \Gamma \frac{1}{2} \\ &= \frac{3}{4} \sqrt{\pi} \end{aligned}$$

$$\begin{aligned} (iv) \Gamma \frac{7}{2} &= \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \Gamma \frac{1}{2} \\ &= \frac{15}{8} \sqrt{\pi} \end{aligned}$$

$$\begin{aligned} (v) \Gamma \frac{9}{2} &= \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \Gamma \frac{1}{2} \\ &= \frac{105}{16} \sqrt{\pi} \end{aligned}$$



$$\begin{aligned} \text{(VI)} \quad \sqrt{-\frac{1}{2}} &= \frac{\sqrt{-\frac{1}{2}+1}}{-\frac{1}{2}} \\ &= \frac{\sqrt{\frac{1}{2}}}{-\frac{1}{2}} \\ &= -2\sqrt{\pi} \end{aligned}$$

$$\begin{aligned} \text{(VII)} \quad \sqrt{-\frac{5}{2}} &= \frac{\sqrt{-\frac{5}{2}+1}}{-\frac{5}{2}} \\ &= \frac{\sqrt{-\frac{3}{2}}}{-\frac{5}{2}} \\ &= \frac{\sqrt{-\frac{3}{2}+1}}{\left(-\frac{5}{2}\right)\left(-\frac{3}{2}\right)} \\ &= \frac{\sqrt{-\frac{1}{2}}}{\left(-\frac{5}{2}\right)\left(-\frac{3}{2}\right)} \end{aligned}$$

$$= \frac{\sqrt{-\frac{1}{2}+1}}{\left(-\frac{5}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{1}{2}\right)} = \frac{\sqrt{\frac{1}{2}}}{-\frac{15}{8}}$$

$$\boxed{\sqrt{-\frac{5}{2}} = -\frac{8}{15}\sqrt{\pi}} \quad n = \frac{\sqrt{n+1}}{n}$$

$$\begin{aligned} \text{(VIII)} \quad \sqrt{-3} &= \frac{\sqrt{-3+1}}{-3} = \frac{\sqrt{-2}}{-3} \\ &= \frac{\sqrt{-2+1}}{(-3)(-2)} = \frac{\sqrt{-1}}{(-3)(-2)} \\ &= \frac{\sqrt{-1+1}}{(-3)(-2)(-1)} = \frac{\sqrt{0}}{-6} \end{aligned}$$

$$\boxed{\sqrt{-3} = \infty}$$

Q.2 Evaluate  $\int_0^{\infty} e^{-x^2} dx$

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$$\int_0^{\infty} e^{-x^2} dx$$

Put  $x^2 = t$

$$2x dx = dt$$

$$dx = \frac{dt}{2x}$$

$$dx = \frac{dt}{2\sqrt{t}}$$

$$dx = \frac{1}{2} t^{-1/2} dt$$

When  $x=0, t=0$

When  $x=\infty, t=\infty$

$$= \int_0^{\infty} e^{-t} \times \frac{1}{2} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t} t^{-1/2} dt$$

$$= \frac{1}{2} \Gamma(-1/2 + 1)$$

$$= \frac{1}{2} \sqrt{\frac{1}{2}}$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}$$



Q.3 Evaluate  $\int_0^{\infty} \sqrt[4]{x} e^{-\sqrt{x}} dx$

$$\int_0^{\infty} \sqrt[4]{x} e^{-\sqrt{x}} dx$$

Put  $\sqrt{x} = t$

$$x = t^2$$

$$dx = 2t dt$$

When  $x = 0$ ,  $t = 0$

When  $x = \infty$ ,  $t = \infty$

$$\int_0^{\infty} (x)^{\frac{1}{4}} e^{-\sqrt{x}} dx = \int_0^{\infty} (t^2)^{\frac{1}{4}} e^{-t} 2t dt$$

$$= 2 \int_0^{\infty} t^{\frac{1}{2}} e^{-t} t dt$$

$$= 2 \int_0^{\infty} e^{-t} t^{\frac{3}{2}} dt$$

$$= 2 \left[ \frac{3}{2} + 1 \right] = 2 \sqrt{\frac{5}{2}}$$

$$= \cancel{2} \times \frac{3}{\cancel{2}} \times \frac{1}{2} \sqrt{\pi} = \boxed{\frac{3}{2} \sqrt{\pi}}$$

$$\int_0^1 x^m (\log x)^n dx$$

OR

$$\int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx$$

Q.4 Evaluate  $\int_0^1 (x \log x)^3 dx$ .

$$\int_0^1 (n \log n)^3 dn$$

$$= \int_0^1 n^3 (\log n)^3 dn$$

Put  $\log n = -t$

$$n = e^{-t}$$

$$dn = -e^{-t} dt$$

When  $n = 0$ ,  $t = \infty$

When  $n = 1$ ,  $t = 0$

$$= \int_{\infty}^0 (e^{-t})^3 (-t)^3 (-e^{-t} dt)$$

$$= \int_{\infty}^0 e^{-3t} t^3 e^{-t} dt$$

$$= \int_{\infty}^0 e^{-4t} t^3 dt$$

$$= - \int_0^{\infty} e^{-4t} t^3 dt$$

$$= - \frac{\sqrt{3+1}}{(4)^4}$$

$$= - \frac{\sqrt{4}}{(4)^4} = - \frac{3 \times 2 \times 1}{4 \times 4 \times 4 \times 4 \times 2}$$

$$= - \frac{3}{128}$$



$$\int_0^1 \frac{dn}{\sqrt{-\log n}} = \sqrt{\pi}$$

LHS

$$\int_0^1 \frac{dn}{\sqrt{-\log n}}$$

$$= \int_0^1 \frac{dn}{\sqrt{\log n^{-1}}}$$

$$= \int_0^1 \frac{dn}{\sqrt{\log \frac{1}{n}}}$$

$$= \int_0^1 \left( \log \frac{1}{n} \right)^{-\frac{1}{2}} dn$$

$$\text{Put } \log \frac{1}{n} = t$$

Q.6 Evaluate  $\int_0^{\infty} \frac{x^c}{c^x} dx$

Gateway Classes



UNIT -4 : DPP - 7

Topic : Beta and gamma Function

**Q.1** Prove that  $\Gamma\left(-\frac{3}{2}\right) = 4/3 \sqrt{\pi}$ .

**Q.2** Evaluate  $\int_0^{\infty} e^{-x^4} dx$

**Q.3** Evaluate  $\int_0^{\infty} x e^{-x^4} dx$

**Q.4** Prove that  $\int_0^1 \left(\log \frac{1}{y}\right)^{n-1} dx = \Gamma n$

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 8

#### Today's target

- BETA AND GAMMA FUNCTION (PART – 2)
- PYQs
- DPP



By Gulshan sir



The Beta Function of  $m$  and  $n$  is defined as

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx = \beta(m, n)$$

$$m > 0, n > 0$$

Conclusion :

$$\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1)$$

Symmetry of Beta Function

$$\beta(m, n) = \beta(n, m)$$

OR

### Another form of Beta Function

$$\int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx = \beta(m, n)$$

**Proof:** We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Put  $x = \frac{1}{1+y}$

$$dx = -\frac{1}{(1+y)^2} dy$$

When  $x=0$ ,  $y=\infty$

When  $x=1$ ,  $y=0$

$$\beta(m, n) = \int_{\infty}^0 \left(\frac{1}{1+y}\right)^{m-1} \left(1 - \frac{1}{1+y}\right)^{n-1} \times \frac{-1}{(1+y)^2} dy$$

$$\beta(m, n) = - \int_{\infty}^0 \frac{1}{(1+y)^{m-1}} \times \left(\frac{1+y-1}{1+y}\right)^{n-1} \times \frac{1}{(1+y)^2} dy$$

$$\beta(m, n) = - \int_{\infty}^0 \frac{1}{(1+y)^{m-1}} \times \frac{y^{n-1}}{(1+y)^{n-1}} \times \frac{1}{(1+y)^2} dy$$

$$\beta(m, n) = - \int_{\infty}^0 \frac{y^{n-1}}{(1+y)^{m-1+n-1+2}} dy$$

$$\beta(m, n) = - \int_{\infty}^0 \frac{y^{n-1}}{(1+y)^{m+n}} dy$$



$$\beta(m, n) = \int_0^{\infty} \frac{y^{n-1}}{(1+y)^{m+n}} dy \quad \left\{ \because \int_a^b f(x) dx = - \int_b^a f(x) dx \right.$$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx \quad \left\{ \because \int_a^b f(x) dx = \int_a^b f(t) dt \right.$$

$$\Rightarrow \beta(n, m) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\left\{ \because \beta(m, n) = \beta(n, m) \right\}$$

$$\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

**Proof:** We know that

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx \quad \text{--- ①}$$

$$\Gamma n = k^n \int_0^{\infty} e^{-kx} x^{n-1} dx \quad \text{--- ②}$$

Replace k by z

$$\Gamma n = z^n \int_0^{\infty} e^{-zx} x^{n-1} dx \quad \text{--- ③}$$

Multiply eq<sup>n</sup> ③ by  $e^{-z}$  and  $z^{m-1}$

$$\Gamma n e^{-z} z^{m-1} = e^{-z} z^{m-1} z^n \int_0^{\infty} e^{-zx} x^{n-1} dx$$

$$\Gamma n e^{-z} z^{m-1} = \int_0^{\infty} e^{-z} e^{-zx} z^{m-1} z^n x^{n-1} dx$$

$$\Gamma n e^{-z} z^{m-1} = \int_0^{\infty} e^{-(1+x)z} z^{m+n-1} x^{n-1} dx$$

Integrate wrt z from 0 to  $\infty$

$$\Gamma n \int_0^{\infty} e^{-z} z^{m-1} dz = \int_0^{\infty} \int_0^{\infty} e^{-(1+x)z} z^{m+n-1} x^{n-1} dz dx$$



$$\Gamma n \Gamma m = \int_0^{\infty} x^{n-1} dx \underbrace{\int_0^{\infty} e^{-(1+x)z} z^{m+n-1} dz}$$

$$\Gamma n \Gamma m = \int_0^{\infty} x^{n-1} dx \times \frac{\Gamma m+n}{(1+x)^{m+n}}$$

$$\Gamma n \Gamma m = \Gamma m+n \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}}$$

$$\Gamma n \Gamma m = \Gamma m+n \beta(m, n) \left\{ \because \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} = \beta(m, n) \right\}$$

$$\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

Hence proved

Using Beta and Gamma function, Prove that

$$\int_0^{\frac{\pi}{2}} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma_{\frac{m+1}{2}} \Gamma_{\frac{n+1}{2}}}{2 \Gamma_{\frac{m+n+2}{2}}}$$

We know that

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Put  $x = \sin^2 \theta$

$$dx = 2 \sin \theta \cos \theta d\theta$$

When  $x=0$ ,  $\theta=0^\circ$

When  $x=1$ ,  $\theta=\frac{\pi}{2}$

$$\beta(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1-\sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta$$

$$\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} (\sin^2 \theta)^{m-1} (\cos^2 \theta)^{n-1} \sin \theta \cos \theta d\theta$$

$$\beta(m, n) = 2 \int_0^{\pi/2} (\sin \theta)^{2m-2} (\cos \theta)^{2n-2} \sin \theta \cos \theta d\theta$$

$$\beta(m, n) = 2 \int_0^{\pi/2} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta$$

$$\int_0^{\pi/2} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta = \frac{1}{2} \beta(m, n)$$



$$\int_0^{\pi/2} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta = \frac{\Gamma m \Gamma n}{2 \Gamma m+n}$$

Put  $m = \frac{m+1}{2}$  and  $n = \frac{n+1}{2}$

$$\int_0^{\pi/2} (\sin \theta)^m (\cos \theta)^n d\theta = \frac{\Gamma \frac{m+1}{2} \Gamma \frac{n+1}{2}}{2 \Gamma \frac{m+n+2}{2}}$$

$$\int_0^{\infty} \frac{x^{n-1}}{1+x} dx = \frac{\pi}{\sin n\pi}$$

Using Beta and Gamma function, Prove that  $\Gamma n \Gamma 1-n = \frac{\pi}{\sin n\pi}$

Proof: We know that

$$\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

using

$$\beta(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$\int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

$$\text{Put } m+n = 1 \Rightarrow m = 1-n$$

$$\int_0^{\infty} \frac{x^{n-1}}{1+x} dx = \frac{\Gamma 1-n \Gamma n}{\Gamma 1}$$

$$\Gamma 1-n \Gamma n = \int_0^{\infty} \frac{x^{n-1}}{1+x} dx$$

$$\Gamma n \Gamma 1-n = \frac{\pi}{\sin n\pi}$$

Special case

$$\text{Put } n = \frac{1}{2}$$

$$\Gamma 1-\frac{1}{2} \Gamma \frac{1}{2} = \frac{\pi}{\sin \frac{\pi}{2}}$$

$$\Gamma \frac{1}{2} \Gamma \frac{1}{2} = \pi$$

$$\left(\Gamma \frac{1}{2}\right)^2 = \pi$$

$$\Gamma \frac{1}{2} = \sqrt{\pi}$$



$$\Gamma m \Gamma m + \frac{1}{2} = \frac{\sqrt{\pi} \Gamma 2m}{2^{2m-1}}$$

**Proof:** We know that

$$\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta$$

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta = \frac{1}{2} \beta(m, n)$$

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta = \frac{\Gamma m \Gamma n}{2 \Gamma m+n}$$

①

$$\text{Put } 2n-1 = 0$$

$$n = \frac{1}{2}$$

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma \frac{1}{2}}{2 \Gamma m + \frac{1}{2}} \quad \text{--- ②}$$

$$\text{Put } n = m \quad \text{①}$$

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{2 \Gamma 2m}$$

$$\int_0^{\frac{\pi}{2}} (\sin \theta \cos \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{2 \Gamma 2m}$$

$$\int_0^{\frac{\pi}{2}} \left( \frac{2 \sin \theta \cos \theta}{2} \right)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{2 \Gamma 2m}$$

$$\frac{1}{2^{2m-1}} \int_0^{\frac{\pi}{2}} (\sin 2\theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{2 \Gamma 2m}$$

Put  $2\theta = t$  | When  $\theta = 0, t = 0$   
 $d\theta = \frac{dt}{2}$  | When  $\theta = \frac{\pi}{2}, t = \pi$

$$\int_0^{\pi} (\sin t)^{2m-1} \frac{dt}{2} = \frac{\Gamma m \Gamma m}{2 \Gamma 2m} \times 2^{2m-1}$$

$$\int_0^{\pi} (\sin \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{\Gamma 2m} 2^{2m-1} \left\{ \because \int_0^a f(x) dx = \int_0^a f(t) dt \right\}$$

$$2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{\Gamma 2m} \times 2^{2m-1} \left\{ \begin{array}{l} \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \\ \text{if } f(2a-x) = f(x) \end{array} \right\}$$

$$\int_0^{\pi/2} (\sin \theta)^{2m-1} d\theta = \frac{\Gamma m \Gamma m}{2 \Gamma 2m} \times 2^{2m-1} \quad \text{--- (3)}$$

Combined (2) and (3)

$$\frac{\Gamma m \Gamma m}{2 \Gamma 2m} \times 2^{2m-1} = \frac{\Gamma m \sqrt{\pi}}{2 \Gamma m + \frac{1}{2}}$$

$$\Gamma m \Gamma m + \frac{1}{2} = \frac{\sqrt{\pi} \Gamma 2m}{2^{2m-1}}$$

Hence proved



Q.1 Evaluate  $\Gamma \frac{1}{4} \Gamma \frac{3}{4}$

AKTU 2024

$$\Gamma \frac{1}{4} \Gamma \frac{3}{4} = \Gamma \frac{1}{4} \Gamma 1 - \frac{1}{4}$$

$$= \frac{\pi}{\sin \frac{\pi}{4}} \left\{ \because \Gamma n \Gamma 1-n = \frac{\pi}{\sin n\pi} \right\}$$

$$= \frac{\pi}{\frac{1}{\sqrt{2}}}$$

$$= \pi \sqrt{2}$$

Q.2 Prove that  $\frac{\Gamma(\frac{1}{3})\Gamma(\frac{5}{6})}{\Gamma(\frac{2}{3})} = (2)^{1/3}\sqrt{\pi}$

AKTU 2017

$$\frac{\Gamma(\frac{1}{3})\Gamma(\frac{5}{6})}{\Gamma(\frac{2}{3})} = (2)^{1/3}\sqrt{\pi}$$

LHS

$$\frac{\Gamma(\frac{1}{3})\Gamma(\frac{5}{6})}{\Gamma(\frac{2}{3})} = \frac{\Gamma(\frac{1}{3})\Gamma(\frac{1}{3} + \frac{1}{2})}{\Gamma(\frac{2}{3})}$$

$$= \frac{\sqrt{\pi} \cancel{\Gamma(\frac{2}{3})}}{\cancel{\Gamma(\frac{2}{3})} 2^{2 \times \frac{1}{3} - 1}} \left\{ \Gamma(m) \Gamma(m + \frac{1}{2}) = \frac{\sqrt{\pi} \Gamma(2m)}{2^{2m-1}} \right\}$$

$$\frac{\Gamma(\frac{1}{3})\Gamma(\frac{5}{6})}{\Gamma(\frac{2}{3})} = \frac{\sqrt{\pi}}{2^{-1/3}}$$

$$\frac{\Gamma(\frac{1}{3})\Gamma(\frac{5}{6})}{\Gamma(\frac{2}{3})} = 2^{1/3}\sqrt{\pi}$$

Hence proved



Q.3 Prove that  $\beta(p, q) = \beta(p+1, q) + \beta(p, q+1)$

AKTU 2024

OR

$$\overline{n+1} = n \overline{n}$$

Prove that  $\beta(m, n) = \beta(m+1, n) + \beta(m, n+1)$

$$\beta(p, q) = \beta(p+1, q) + \beta(p, q+1)$$

$$= \frac{p \overline{p} \overline{q}}{(p+q) \overline{p+q}} + \frac{\overline{p} q \overline{q}}{(p+q) \overline{p+q}} \quad \left\{ \because \overline{n+1} = n \overline{n} \right.$$

RHS

$$\beta(p+1, q) + \beta(p, q+1)$$

$$= \frac{\overline{p} \overline{q}}{\overline{p+q}} \left( \frac{p}{p+q} + \frac{q}{p+q} \right)$$

$$\text{using } \beta(m, n) = \frac{\overline{m} \overline{n}}{\overline{m+n}}$$

$$= \frac{\overline{p} \overline{q}}{\overline{p+q}} \left( \frac{\cancel{p+q}}{\cancel{p+q}} \right)$$

$$= \frac{\overline{p+1} \overline{q}}{\overline{p+q+1}} + \frac{\overline{p} \overline{q+1}}{\overline{p+q+1}}$$

$$= \beta(p, q) \Rightarrow \text{LHS} = \text{RHS}$$

Q.4 Evaluate  $\int_0^{\infty} \frac{x^4(1+x^5)}{(1+x)^{15}} dx$

$$\int_0^{\infty} \frac{x^4(1+x^5)}{(1+x)^{15}} dx$$

$$= \int_0^{\infty} \frac{x^4 + x^9}{(1+x)^{15}} dx$$

$$= \int_0^{\infty} \frac{x^4}{(1+x)^{15}} dx + \int_0^{\infty} \frac{x^9}{(1+x)^{15}} dx$$

$$\beta(m, n) = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^{m+n}} dx$$

$$= \int_0^{\infty} \frac{x^{5-1}}{(1+x)^{10+5}} dx + \int_0^{\infty} \frac{x^{10-1}}{(1+x)^{5+10}} dx$$

$$= \beta(10, 5) + \beta(5, 10)$$

$$= \beta(10, 5) + \beta(10, 5) \quad \left\{ \beta(m, n) = \beta(n, m) \right\}$$

$$= 2\beta(10, 5) \quad \left\{ \because \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \right\}$$

$$= \frac{2 \Gamma(10) \Gamma(5)}{\Gamma(10+5)} = \frac{2 \times 9! \times 4!}{14!}$$

$$= \frac{1}{5005} \text{ Ans}$$



UNIT -4 : DPP - 8

Topic : Beta and gamma Function

Q.1 Prove that  $\beta(m, n) = \beta(n, m)$

Q.2 Prove that  $\frac{B(p, q+1)}{q} = \frac{B(p+1, q)}{p} = \frac{B(p, q)}{p+q}$ ,  $(p > 0, q > 0)$

AKTU 2015

Q.3 Evaluate  $\int_0^{\infty} \frac{x^8(1-x^6)}{(1+x)^{24}} dx$

Q.4 Assuming  $\Gamma n \Gamma 1-n = \pi \operatorname{cosec} n\pi$ ,  $0 < n < 1$ , show that  $\int_0^{\infty} \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin n\pi}$ ;  $0 < p < 1$

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 9

#### Today's target

- BETA AND GAMMA FUNCTION (PART – 3)
- PYQs
- DPP



By Gulshan sir



$$(1) \int_0^{\frac{\pi}{2}} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma_{\frac{m+1}{2}} \Gamma_{\frac{n+1}{2}}}{2 \Gamma_{\frac{m+n+2}{2}}}$$

(2) If  $n$  is a positive integer (Natural Number)

$$\Gamma n = (n-1)!$$

(3) If  $n$  is a positive fraction (Numerator > Denominator)

$$\Gamma n = (n-1) \Gamma_{n-1}$$

$$(4) \Gamma n \Gamma_{1-n} = \frac{\pi}{\sin n \pi} \quad (\text{Numerator} < \text{Denominator})$$

$$(5) \text{ Duplication formula : } \Gamma m \Gamma_{m+\frac{1}{2}} = \frac{\sqrt{\pi} \Gamma_{2m}}{2^{2m-1}} \quad (\text{Numerator} < \text{Denominator})$$

### Important Results

$$(i) \Gamma 1 = 1$$

$$(ii) \Gamma_{\frac{1}{2}} = \sqrt{\pi}$$

$$(iii) \Gamma 0 = \infty$$

Q.1 Prove the following result  $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{\pi}{\sqrt{2}}$

AKTU 2019

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta = \int_0^{\frac{\pi}{2}} \sqrt{\cot \theta} d\theta = \frac{\pi}{\sqrt{2}}$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos \theta}}{\sqrt{\sin \theta}} d\theta$$

$$= \int_0^{\frac{\pi}{2}} (\sin \theta)^{-1/2} (\cos \theta)^{1/2} d\theta$$

Using

$$\int_0^{\pi/2} (\sin \theta)^m \times (\cos \theta)^n d\theta = \frac{\left[\frac{m+1}{2}\right] \left[\frac{n+1}{2}\right]}{2 \left[\frac{m+n+2}{2}\right]}$$

$$\int_0^{\pi/2} (\sin \theta)^{-1/2} (\cos \theta)^{1/2} d\theta = \frac{\left[\frac{-1/2+1}{2}\right] \left[\frac{1/2+1}{2}\right]}{2 \left[\frac{-1/2+1/2+2}{2}\right]}$$

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{\tan(\frac{\pi}{2} - \theta)} d\theta \left\{ \int_0^a f(x) dx = \int_0^a f(a-x) dx \right\}$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{\cot \theta} d\theta$$



$$= \frac{\sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}}}{2 \sqrt{1}}$$

$$= \frac{\sqrt{\frac{1}{4}} \sqrt{1 - \frac{1}{4}}}{2}$$

$$= \frac{\pi}{2 \sin \frac{\pi}{4}}$$

$$= \frac{\pi}{2 \times \frac{1}{\sqrt{2}}}$$

$$\left\{ \begin{aligned} \pi n \sqrt{1-n} &= \frac{\pi}{\sin n\pi} \end{aligned} \right.$$

$$= \frac{\pi}{\frac{2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}}$$

$$= \frac{\pi}{\sqrt{2}}$$

Hence proved

Q.2 Prove the following result  $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$

AKTU 2014

$$\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$$

LHS

$$\int_0^{\frac{\pi}{2}} \frac{d\theta}{(\sin \theta)^{1/2}} \times \int_0^{\frac{\pi}{2}} (\sin \theta)^{1/2} d\theta$$

$$= \int_0^{\frac{\pi}{2}} (\sin \theta)^{-1/2} (\cos \theta)^0 \times \int_0^{\frac{\pi}{2}} (\sin \theta)^{1/2} (\cos \theta)^0 d\theta$$

using

$$\int_0^{\pi/2} (\sin \theta)^m (\cos \theta)^n d\theta = \frac{\left| \frac{m+1}{2} \right| \left| \frac{n+1}{2} \right|}{2 \left| \frac{m+n+2}{2} \right|}$$

$$= \frac{\left| \frac{-\frac{1}{2}+1}{2} \right| \left| \frac{0+1}{2} \right|}{2 \left| \frac{-\frac{1}{2}+0+2}{2} \right|} \times \frac{\left| \frac{\frac{1}{2}+1}{2} \right| \left| \frac{0+1}{2} \right|}{2 \left| \frac{\frac{1}{2}+0+2}{2} \right|}$$

$$= \frac{\left| \frac{1}{4} \right| \left| \frac{1}{2} \right|}{2 \left| \frac{3}{4} \right|} \times \frac{\left| \frac{3}{4} \right| \left| \frac{1}{2} \right|}{2 \left| \frac{5}{4} \right|}$$

$$= \frac{\cancel{\frac{1}{4}} \left| \frac{1}{2} \right| \cancel{\frac{1}{2}}}{\cancel{4} \times \frac{1}{\cancel{4}} \cancel{\left| \frac{1}{4} \right|}}$$

$$= \sqrt{\pi} \times \sqrt{\pi} = \pi$$

Hence proved



Q.3 Evaluate  $\int_0^{\infty} \frac{dx}{1+x^4}$

$$\int_0^{\infty} \frac{dx}{1+x^4}$$

Put  $x^4 = \tan^2 \theta$

$$x = (\tan \theta)^{1/2}$$

$$dx = \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta d\theta$$

When  $x = 0$ ,  $\theta = 0^\circ$

When  $x = \infty$ ,  $\theta = \frac{\pi}{2}$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{(\tan \theta)^{-1/2} \sec^2 \theta d\theta}{1 + \tan^2 \theta} \quad \text{AKTU 2019}$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{(\tan \theta)^{-1/2} \cancel{\sec^2 \theta} d\theta}{\cancel{\sec^2 \theta}}$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{(\sin \theta)^{-1/2}}{(\cos \theta)^{-1/2}}$$

$$= \frac{1}{2} \int_0^{\pi/2} (\sin \theta)^{-1/2} (\cos \theta)^{1/2} d\theta$$

using

$$\int_0^{\frac{\pi}{2}} (\sin \theta)^m (\cos \theta)^n d\theta = \frac{\left[\frac{m+1}{2}\right] \left[\frac{n+1}{2}\right]}{2 \left[\frac{m+n+2}{2}\right]}$$

$$= \frac{\left[\frac{-\frac{1}{2}+1}{2}\right] \left[\frac{\frac{1}{2}+1}{2}\right]}{2 \times 2 \left[\frac{-\frac{1}{2}+\frac{1}{2}+2}{2}\right]}$$

$$= \frac{\left[\frac{1}{4}\right] \left[\frac{3}{4}\right]}{4 \left[\frac{1}{1}\right]}$$

$$= \frac{\left[\frac{1}{4}\right] \left[\frac{3}{4}\right]}{4}$$

$$= \frac{\pi}{4 \sin \frac{\pi}{4}}$$

$$\left[\frac{n}{2}\right] \left[\frac{1-n}{2}\right] = \frac{\pi}{\sin n\pi}$$

$$= \frac{\pi \sqrt{2}}{4}$$



Q.4 Evaluate  $\int_0^1 \frac{dx}{\sqrt{1+x^4}}$

$$\int_0^1 \frac{dx}{\sqrt{1+x^4}}$$

Put  $x^4 = \tan^2 \theta$

$$x = (\tan \theta)^{1/2}$$

$$dx = \frac{1}{2} (\tan \theta)^{-1/2} \sec^2 \theta d\theta$$

When  $x=0$ ,  $\theta=0$

When  $x=1$ ,  $\theta = \frac{\pi}{4}$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{(\tan \theta)^{-1/2} \sec^2 \theta d\theta}{\sqrt{1+\tan^2 \theta}}$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{(\tan \theta)^{-1/2} \sec^2 \theta d\theta}{\sec \theta}$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{(\sin \theta)^{-1/2}}{(\cos \theta)^{-1/2}} \times \frac{1}{\cos \theta} d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} (\sin \theta)^{-1/2} (\cos \theta)^{-1/2} d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} (\sin \theta \cos \theta)^{-1/2} d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} \left( \frac{2 \sin \theta \cos \theta}{2} \right)^{-1/2} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/4} \left( \frac{\sin 2\theta}{2} \right)^{-1/2} d\theta$$

$$= \frac{1}{2 \times 2^{-1/2}} \int_0^{\pi/4} (\sin 2\theta)^{-1/2} d\theta$$

Put  $2\theta = t$

$$d\theta = \frac{dt}{2}$$

When  $\theta = 0, t = 0$

When  $\theta = \frac{\pi}{4}, t = \frac{\pi}{2}$

$$= \frac{\sqrt{2}}{2} \int_0^{\pi/2} (\sin t)^{-1/2} \frac{dt}{2}$$

$$= \frac{\sqrt{2}}{4} \int_0^{\pi/2} (\sin t)^{-1/2} (\cos t)^0 dt$$

$$= \frac{\sqrt{2}}{4} \int_0^{\pi/2} (\sin \theta)^{-1/2} (\cos \theta)^0 d\theta \left\{ \int_a^b f(n) dn = \int_a^b f(t) dt \right\}$$

$$= \frac{\sqrt{2} \left| \frac{-\frac{1}{2}+1}{2} \right| \left| \frac{0+1}{2} \right|}{4 \times 2 \left| \frac{-\frac{1}{2}+0+2}{2} \right|}$$

$$= \frac{\sqrt{2} \left| \frac{1}{4} \right| \left| \frac{1}{2} \right|}{8 \left| \frac{3}{4} \right|}$$

$$= \frac{\sqrt{2} \sqrt{\frac{1}{4}}}{8 \sqrt{\frac{3}{4}}}$$

$$= \frac{\sqrt{2} \sqrt{\frac{1}{4}} \sqrt{\frac{1}{4}}}{8 \sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}}}$$

$$= \frac{\sqrt{2} \left( \sqrt{\frac{1}{4}} \right)^2}{8 \sqrt{\frac{1}{4}} \sqrt{1-\frac{1}{4}}}$$

$$= \frac{\sqrt{2} \left( \sqrt{\frac{1}{4}} \right)^2}{8 \frac{\pi}{\sin \frac{\pi}{4}}} \left\{ \because \sqrt{n} \sqrt{1-n} = \frac{\pi}{\sin n\pi} \right\}$$

$$= \frac{1}{8\sqrt{2}} \left( \sqrt{\frac{1}{4}} \right)^2 \text{ Ans}$$



Q.5 Using beta gamma functions, Evaluate  $\int_0^1 \left(\frac{x^3}{1-x^3}\right)^{\frac{1}{2}} dx$

$$\int_0^1 \left(\frac{x^3}{1-x^3}\right)^{\frac{1}{2}} dx$$

Put  $x^3 = \sin^2 \theta$

$$x = (\sin \theta)^{2/3}$$

$$dx = \frac{2}{3} (\sin \theta)^{-1/3} \cos \theta$$

When  $x=0$ ,  $\theta=0$

When  $x=1$ ,  $\theta = \frac{\pi}{2}$

$$= \int_0^{\pi/2} \left(\frac{\sin^2 \theta}{1-\sin^2 \theta}\right)^{\frac{1}{2}} \times \frac{2}{3} (\sin \theta)^{-1/3} \cos \theta d\theta$$

$$= \frac{2}{3} \int_0^{\pi/2} \left(\frac{\sin^2 \theta}{\cos^2 \theta}\right)^{\frac{1}{2}} (\sin \theta)^{-1/3} \cos \theta d\theta$$

$$= \frac{2}{3} \int_0^{\pi/2} \frac{\sin \theta}{\cos \theta} \times (\sin \theta)^{-1/3} \cos \theta d\theta$$

$$= \frac{2}{3} \int_0^{\pi/2} (\sin \theta)^{2/3} (\cos \theta)^0 d\theta$$

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$$= \frac{\cancel{2} \sqrt{\frac{2}{3}+1} \sqrt{\frac{0+1}{2}}}{3 \times \cancel{2} \sqrt{\frac{2}{3}+0+2}}$$

$$= \frac{\sqrt{\frac{5}{6}} \sqrt{\frac{1}{2}}}{3 \sqrt{\frac{8}{6}}}$$

$$= \frac{\sqrt{\frac{5}{6}} \sqrt{\pi}}{3 \sqrt{\frac{4}{3}}}$$

$$= \frac{\sqrt{\pi} \sqrt{\frac{5}{6}}}{3\left(\frac{4}{3}-1\right)\sqrt{\frac{4}{3}-1}}$$

$$= \frac{\sqrt{\pi} \sqrt{\frac{5}{6}}}{\cancel{3} \times \frac{1}{\cancel{3}} \sqrt{\frac{1}{3}}}$$

$$= \frac{\sqrt{\pi} \sqrt{\frac{5}{6}}}{\sqrt{\frac{1}{3}}}$$

Ans

Gateway Classes



Q.6 Using beta gamma functions, Evaluate  $\int_0^2 x (8 - x^3)^{\frac{1}{3}} dx$ .

$$\int_0^2 x (8 - x^3)^{\frac{1}{3}} dx$$

Put  $x^3 = 8 \sin^2 \theta$

$$x = 2 (\sin \theta)^{\frac{2}{3}}$$

$$dx = 2 \times \frac{2}{3} (\sin \theta)^{-\frac{1}{3}} \cos \theta d\theta$$

When  $x = 0$ ,  $\theta = 0^\circ$

When  $x = 2$ ,  $\theta = \frac{\pi}{2}$

$$= \int_0^{\pi/2} 2 (\sin \theta)^{\frac{2}{3}} (8 - 8 \sin^2 \theta)^{\frac{1}{3}} \times \frac{4}{3} (\sin \theta)^{-\frac{1}{3}} \cos \theta d\theta$$

$$= \frac{8}{3} \int_0^{\pi/2} (\sin \theta)^{\frac{2}{3}} (8)^{\frac{1}{3}} (1 - \sin^2 \theta)^{\frac{1}{3}} (\sin \theta)^{-\frac{1}{3}} \cos \theta d\theta$$

$$= \frac{16}{3} \int_0^{\pi/2} (\sin \theta)^{\frac{2}{3}} (\cos \theta)^{\frac{2}{3}} (\sin \theta)^{-\frac{1}{3}} \cos \theta d\theta$$

$$= \frac{16}{3} \int_0^{\pi/2} (\sin \theta)^{\frac{1}{3}} (\cos \theta)^{\frac{5}{3}} d\theta$$

$$= \frac{16 \sqrt{\frac{\frac{1}{3}+1}{2}} \sqrt{\frac{\frac{5}{3}+1}{2}}}{3 \times 2 \sqrt{\frac{\frac{1}{3} + \frac{5}{3} + 2}{2}}}$$

$$= \frac{8 \sqrt{\frac{4}{6}} \sqrt{\frac{8}{6}}}{3 \times 2 \sqrt{2}}$$

$$= \frac{8 \sqrt{\frac{2}{3}} \sqrt{\frac{4}{3}}}{3 \times 1}$$

$$= \frac{8 \sqrt{\frac{2}{3}} \frac{1}{3} \sqrt{\frac{1}{3}}}{3}$$

$$= \frac{8 \sqrt{\frac{2}{3}} \sqrt{\frac{1}{3}}}{9}$$

$$= \frac{8 \sqrt{\frac{1}{3}} \sqrt{\frac{2}{3}}}{9}$$

$$= \frac{8 \sqrt{\frac{1}{3}} \sqrt{1-\frac{1}{3}}}{9}$$

$$= \frac{8 \pi}{9 \sin \frac{\pi}{3}}$$

$$= \frac{16 \pi}{9 \sqrt{3}}$$

$$\left\{ \sqrt{n} \sqrt{1-n} = \frac{\pi}{\sin n \pi} \right\}$$



$$(1) \int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma n$$

$$(2) \int_0^{\infty} e^{-kx} x^{n-1} dx = \frac{\Gamma n}{k^n}$$

$$(3) \Gamma n + 1 = n \Gamma n$$

$$(4) \Gamma n + 1 = n!$$

## Important Results

$$(i) \Gamma 1 = 1$$

$$(ii) \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$(iii) \Gamma 0 = \infty$$

## NOTE :

(1) If  $n$  is a positive integer (Natural Number)

$$\Gamma n = (n - 1)!$$

(2) If  $n$  is a positive fraction (Numerator > Denominator)

$$\Gamma n = (n - 1) \Gamma n - 1$$

(3) If  $n$  is a negative fraction

$$\Gamma n = \frac{\Gamma n+1}{n}$$

(4) If  $n$  is a negative integer

$$\Gamma n = \frac{\Gamma n+1}{n}$$

Note :  $\Gamma n = \infty$

$$(1) \int_0^1 x^{m-1} (1-x)^{n-1} dx = \beta(m, n)$$

$$(2) \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx = \beta(m, n)$$

$$(3) \beta(m, n) = \beta(n, m)$$

$$(4) \beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

$$(5) \int_0^{\frac{\pi}{2}} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma \frac{m+1}{2} \Gamma \frac{n+1}{2}}{2 \Gamma \frac{m+n+2}{2}}$$

$$(6) \Gamma n \Gamma 1-n = \frac{\pi}{\sin n \pi} \text{ (Numerator < Denominator)}$$

$$(7) \text{ Duplication formula : } \Gamma m \Gamma m + \frac{1}{2} = \frac{\sqrt{\pi} \Gamma 2m}{2^{2m-1}} \text{ (Numerator < Denominator)}$$



UNIT -4 : DPP - 9

Topic : BETA AND GAMMA FUNCTION

Q.1 Prove that  $\int_0^{\pi/2} \tan^n x \, dx = \frac{\pi}{2} \sec \frac{n\pi}{2}$

Q.2 Using beta gamma functions, Evaluate  $\int_0^1 x^5 (1 - x^3)^{10} \, dx$ .

Q.3 Evaluate  $\int_0^1 x^4 (1 - \sqrt{x})^5 \, dx$

Q.4 Prove that  $\beta(m, n) = a^m b^n \int_0^\infty \frac{x^{m-1}}{(ax+b)^{m+n}} \, dx$

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Q.5 Use beta gamma functions to solve  $\int_0^\infty \frac{dx}{1+x^4} \int_0^{\pi/2} \sqrt{\cot \theta} \, d\theta$

AKTU 2019

## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 10

#### Today's target

- BETA AND GAMMA FUNCTION : Dirichlet's Integral
- PYQs
- DPP



By Gulshan sir



First octant

If  $v$  is the region bounded by  $x \geq 0, y \geq 0, z \geq 0$  and subject to the condition  $x + y + z \leq 1$ , then

Dirichlet's Integral is

$$\iiint_v x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\Gamma l \Gamma m \Gamma n}{\Gamma l + m + n + 1}$$

$\Gamma n$

OR

If  $v$  is the region bounded by  $u \geq 0, v \geq 0, w \geq 0$  and subject to the condition  $u + v + w \leq 1$ , then

Dirichlet's Integral is

$$\iiint_v u^{l-1} v^{m-1} w^{n-1} du dv dw = \frac{\Gamma l \Gamma m \Gamma n}{\Gamma l + m + n + 1}$$

**Note : Application of Dirichlet's Integral**

$$\text{Volume} = \iiint dx dy dz$$

$$\text{Mass} = \iiint \rho dx dy dz$$

$$\textcircled{1} \Gamma n = (n-1)!$$

$$\textcircled{2} \Gamma n = (n-1) \Gamma n-1$$

**Q.1** Evaluate  $\iiint_v dx dy dz$ , where  $v$  is defined by  $x \geq 0, y \geq 0, z \geq 0$  and  $x + y + z \leq 1$

Given

$$x \geq 0, y \geq 0, z \geq 0 \text{ and } x + y + z \leq 1$$

$$\iiint_v dx dy dz = \iiint_v x^0 y^0 z^0 dx dy dz$$

$$= \iiint_v x^{1-1} y^{1-1} z^{1-1} dx dy dz$$

$$= \frac{\pi \pi \pi}{\sqrt{1+1+1+1}} \left\{ \begin{array}{l} \text{If } x \geq 0, y \geq 0, z \geq 0 \text{ and } x+y+z \leq 1 \\ \text{then } \iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\pi^l \pi^m \pi^n}{\sqrt{l+m+n+1}} \end{array} \right\}$$

$$= \frac{1}{\sqrt{4}}$$

$$= \frac{1}{3!}$$

$$= \frac{1}{3 \times 2 \times 1}$$

$$= \frac{1}{6}$$



Q.2 Use Dirichlet's Integral to evaluate  $\iiint xyz \, dx dy dz$ , through out the volume bounded

by  $x = 0, y = 0, z = 0$  where  $x + y + z = 1$

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Given

$x = 0, y = 0, z = 0$  and  $x + y + z = 1$

$$\iiint xyz \, dx dy dz = \iiint x^{2-1} y^{2-1} z^{2-1} \, dx dy dz$$

$$= \frac{\sqrt{2} \sqrt{2} \sqrt{2}}{\sqrt{2+2+2+1}} \left\{ I \right\}$$

$$= \frac{1 \times 1 \times 1}{\sqrt{7}}$$

$$= \frac{1}{6!}$$

$$= \frac{1}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= \frac{1}{720}$$

**Q.3** The plane  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$  meets the axes in A, B and C. Apply Dirichlet's integral to find the volume of the tetrahedron OABC. Also find its mass if the density at any point is  $kxyz$

Given

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$x \geq 0, y \geq 0, z \geq 0$  and  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$  Volume of tetrahedron

Put  $\frac{x}{a} = u$        $\frac{y}{b} = v$        $\frac{z}{c} = w$

$x = au$        $y = bv$        $z = cw$

$dx = a du$        $dy = b dv$        $dz = c dw$

Then

$u > 0, v > 0, w > 0$  and  $u + v + w = 1$

$$V = \iiint dx dy dz$$

$$V = \iiint a du \times b dv \times c dw$$

$$V = abc \iiint du dv dw$$



$$V = abc \iiint u^0 v^0 w^0 du dv dw$$

$$= abc \iiint u^{1-1} v^{1-1} w^{1-1} du dv dw$$

$$= \frac{abc \times \pi \pi \pi}{1+1+1+1} \left\{ I \right\} -$$

$$= \frac{abc}{14}$$

$$= \frac{abc}{31}$$

$$= \frac{abc}{6}$$

$$\text{Mass} = \iiint \rho dx dy dz$$

$$= \iiint kxyz dx dy dz$$

$$= k \iiint a x b v x c w x a d u x b d v x c d w$$

$$= k a^2 b^2 c^2 \iiint u v w du dv dw$$

$$= k a^2 b^2 c^2 \iiint u^{2-1} v^{2-1} w^{2-1} du dv dw$$

$$= k a^2 b^2 c^2 \times \frac{\sqrt{2} \sqrt{2} \sqrt{2}}{\sqrt{2+2+2+1}}$$

$$\text{Mass} = \frac{K a^2 b^2 c^2 \times 1! \times 1! \times 1!}{7!}$$

$$= \frac{K a^2 b^2 c^2 \times 1 \times 1 \times 1}{6!}$$

$$= \frac{K a^2 b^2 c^2}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= \frac{K a^2 b^2 c^2}{720}$$



first octant of the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  if density at any point is that  $\rho(x, y, z) = kxyz$

Given

$$x > 0, y > 0, z > 0 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Put  $\frac{x^2}{a^2} = u$

$$x^2 = a^2 u$$

$$x = a\sqrt{u}$$

$$dx = \frac{a}{2\sqrt{u}} du$$

$$\frac{y^2}{b^2} = v$$

$$y^2 = b^2 v$$

$$y = b\sqrt{v}$$

$$dy = \frac{b}{2\sqrt{v}} dv$$

$$\frac{z^2}{c^2} = w$$

$$z^2 = c^2 w$$

$$z = c\sqrt{w}$$

$$dz = \frac{c}{2\sqrt{w}} dw$$

Then

$$u > 0, v > 0, w > 0 \text{ and } u + v + w = 1$$

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Volume in first octant

$$V = \iiint dx dy dz$$

$$V = \iiint \frac{a du}{2\sqrt{u}} \times \frac{b dv}{2\sqrt{v}} \times \frac{c dw}{2\sqrt{w}}$$

$$V = \frac{abc}{2 \times 2 \times 2} \iiint \frac{1}{\sqrt{u}} \times \frac{1}{\sqrt{v}} \times \frac{1}{\sqrt{w}} du dv dw$$

$$V = \frac{abc}{8} \iiint u^{-1/2} v^{-1/2} w^{-1/2} du dv dw$$

$$V = \frac{abc}{8} \iiint u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1} du dv dw$$

$$V = \frac{abc \times \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \times \sqrt{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + 1}}$$

$$V = \frac{abc \times \sqrt{\pi} \times \sqrt{\pi} \times \sqrt{\pi}}{8 \sqrt{\frac{5}{2}}}$$

$$V = \frac{\pi \sqrt{\pi} abc}{2 \cdot 8 \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}$$

$$V = \frac{\pi abc}{6}$$

$$\text{Mass} = \iiint \rho dn dy dz$$

$$\text{Mass} = \iiint k n y z dn dy dz$$

$$= k \iiint a \sqrt{u} \times b \sqrt{v} \times c \sqrt{w} \times \frac{a du}{2\sqrt{u}} \times \frac{b dv}{2\sqrt{v}} \times \frac{c dw}{2\sqrt{w}}$$

$$= \frac{k a^2 b^2 c^2}{8} \iiint du dv dw$$

$$= \frac{k a^2 b^2 c^2}{8} \iiint u^{1-1} v^{1-1} w^{1-1} du dv dw$$



$$\text{Mass} = \frac{K a^2 b^2 c^2}{8} \times \frac{\pi \pi \pi}{\sqrt{1+1+1+1}} \left\{ I \right\}$$

$$= \frac{K a^2 b^2 c^2}{8} \times \frac{1}{14}$$

$$= \frac{K a^2 b^2 c^2}{8 \times 3!}$$

$$= \frac{K a^2 b^2 c^2}{48}$$

Q.5 Find the volume of the solid bounded by the co-ordinate planes and the surface

$$\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} + \sqrt{\frac{z}{c}} = 1$$

AKTU 2024

Let

$$x \geq 0, y \geq 0, z \geq 0 \text{ and } \sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} + \sqrt{\frac{z}{c}} = 1$$

Put  $\sqrt{\frac{x}{a}} = u$

$$\frac{x}{a} = u^2$$

$$x = au^2$$

$$dx = 2au du$$

$$\sqrt{\frac{y}{b}} = v$$

$$\frac{y}{b} = v^2$$

$$y = bv^2$$

$$dy = 2bv dv$$

$$\sqrt{\frac{z}{c}} = w$$

$$\frac{z}{c} = w^2$$

$$z = cw^2$$

$$dz = 2cw dw$$

Then

$$u > 0, v > 0, w > 0 \text{ and } u + v + w = 1$$

Volume of solid in first octant

$$V = \iiint dx dy dz$$

$$V = \iiint 2a u du \times 2b v dv \times 2c w dw$$

$$V = 8abc \iiint u v w du dv dw$$

$$V = 8abc \iiint u^{2-1} v^{2-1} w^{2-1} du dv dw$$



$$V = 8abc \times \frac{\sqrt{2} \sqrt{2} \sqrt{2}}{\sqrt{2+2+2+1}}$$

$$V = \frac{8abc \times 1 \times 1 \times 1}{\sqrt{7}}$$

$$V = \frac{8abc}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$V = \frac{abc}{90}$$

Total volume of the solid

$V = 8 \times \text{volume of First octant}$

$$V = \frac{8abc}{90}$$

Q.6 Apply Dirichlet's integral to find the volume of an octant of sphere  $x^2 + y^2 + z^2 \leq 25$

Let

$$x > 0, y > 0, z > 0 \text{ and } x^2 + y^2 + z^2 = 25$$

OR

$$\frac{x^2}{25} + \frac{y^2}{25} + \frac{z^2}{25} = 1$$

Put

$$\frac{x^2}{25} = u$$

$$x^2 = 25u$$

$$x = 5\sqrt{u}$$

$$dx = \frac{5}{2\sqrt{u}} du$$

$$\frac{y^2}{25} = v$$

$$y^2 = 25v$$

$$y = 5\sqrt{v}$$

$$dy = \frac{5}{2\sqrt{v}} dv$$

$$\frac{z^2}{25} = w$$

$$z^2 = 25w$$

$$z = 5\sqrt{w}$$

$$dz = \frac{5}{2\sqrt{w}} dw$$

Then

$$u > 0, v > 0, w > 0 \text{ and } u + v + w = 1$$

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Volume of sphere in first octant

$$V = \iiint dx dy dz$$

$$V = \iiint \frac{5 du}{2\sqrt{u}} \times \frac{5 dv}{2\sqrt{v}} \times \frac{5 dw}{2\sqrt{w}}$$

$$V = \frac{125}{8} \iiint \frac{1}{\sqrt{u}} \times \frac{1}{\sqrt{v}} \times \frac{1}{\sqrt{w}} du dv dw$$

$$V = \frac{125}{8} \iiint u^{-1/2} v^{-1/2} w^{-1/2} du dv dw$$



$$V = \frac{125}{8} \iiint u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1} du dv dw$$

$$= \frac{125 \times \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \times \left| \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + 1 \right|}$$

$$= \frac{125 \times \sqrt{\pi} \times \sqrt{\pi} \times \sqrt{\pi}}{8 \times \left| \frac{5}{2} \right|}$$

$$= \frac{125 \times \sqrt{\pi} \times \sqrt{\pi} \times \sqrt{\pi}}{2 \times 8 \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}$$

$$V = \frac{125\pi}{6}$$

Q.7 Compute  $\iiint_V x^2 dx dy dz$  over the volume of tetrahedron bounded by  $x=0$ ,  $y=0$ ,  $z=0$

and  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

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Given

$x > 0, y > 0, z > 0$  and  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Put

$\frac{x}{a} = u$

$\frac{y}{b} = v$

$\frac{z}{c} = w$

$x = au$

$y = bv$

$z = cw$

$dx = a du$

$dy = b dv$

$dz = c dw$

Then

$u > 0, v > 0, w > 0$  and  $u + v + w = 1$

$$\iiint_V x^2 dx dy dz$$

$$= \iiint_V a^2 u^2 \cdot a du \cdot b dv \cdot c dw$$

$$= a^3 bc \iiint_V u^2 du dv dw$$

$$= a^3 bc \iiint_V u^{3-1} v^{1-1} w^{1-1} du dv dw$$



$$\iiint x^2 dx dy dz = \frac{a^3 bc \times \sqrt{3} \pi \pi}{\sqrt{3+1+1+1}} \left\{ I \right\} \text{---}$$

$$= \frac{a^3 bc \times 2! \times 1 \times 1}{\sqrt{6}}$$

$$= \frac{a^3 bc \times 2}{5!}$$

$$= \frac{a^3 bc \times \cancel{2} \times 1}{5 \times 4 \times 3 \times 2 \times \cancel{1}}$$

$$= \boxed{\frac{a^3 bc}{60}}$$

**GW** Q.8 Evaluate  $\iiint (ax^2 + by^2 + cz^2) dx dy dz$ , where  $v$  is the region bounded by  $x^2 + y^2 + z^2 \leq 1$  **GW**

Let  $x \geq 0, y \geq 0, z \geq 0$  and  $x^2 + y^2 + z^2 \leq 1$

$$= \iiint ax^2 dx dy dz + \iiint by^2 dx dy dz + \iiint cz^2 dx dy dz$$

Let  $x^2 = u \quad y^2 = v \quad z^2 = w$

$$x = \sqrt{u} \quad y = \sqrt{v} \quad z = \sqrt{w}$$

$$dx = \frac{1}{2\sqrt{u}} du \quad dy = \frac{1}{2\sqrt{v}} dv \quad dz = \frac{1}{2\sqrt{w}} dw$$

$$= a \iiint \frac{u}{2\sqrt{u}} \frac{dv}{2\sqrt{v}} \frac{dw}{2\sqrt{w}} + b \iiint \frac{v}{2\sqrt{u}} \frac{dv}{2\sqrt{v}} \frac{dw}{2\sqrt{w}} + c \iiint \frac{w}{2\sqrt{u}} \frac{dv}{2\sqrt{v}} \frac{dw}{2\sqrt{w}}$$

Then  $u \geq 0, v \geq 0, w \geq 0$  and  $u + v + w = 1$

$$\iiint (ax^2 + by^2 + cz^2) dx dy dz$$

$$= \frac{a}{8} \iiint u^{1/2} v^{-1/2} w^{-1/2} du dv dw + \frac{b}{8} \iiint u^{-1/2} v^{1/2} w^{-1/2} du dv dw$$

$$+ \frac{c}{8} \iiint u^{-1/2} v^{-1/2} w^{1/2} du dv dw$$



$$= \frac{a}{8} \iiint u^{\frac{3}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1} du dv dw + \frac{b}{8} \iiint u^{\frac{1}{2}-1} v^{\frac{3}{2}-1} w^{\frac{1}{2}-1} du dv dw + \frac{c}{8} \iiint u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{3}{2}-1} du dv dw$$

$$= \frac{a \sqrt{\frac{3}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \left| \frac{3}{2} + \frac{1}{2} + \frac{1}{2} + 1 \right|} + \frac{b \sqrt{\frac{1}{2}} \sqrt{\frac{3}{2}} \sqrt{\frac{1}{2}}}{8 \left| \frac{1}{2} + \frac{3}{2} + \frac{1}{2} + 1 \right|} + \frac{c \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{3}{2}}}{8 \left| \frac{1}{2} + \frac{1}{2} + \frac{3}{2} + 1 \right|} \left\{ \begin{array}{l} \text{If } u > 0, v > 0, w > 0 \text{ and } u+v+w=1 \\ \text{Then } \iiint x^{l-1} y^{m-1} z^{n-1} du dv dw = \frac{\Gamma(l) \Gamma(m) \Gamma(n)}{\Gamma(l+m+n+1)} \end{array} \right.$$

$$= \frac{a \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \sqrt{\frac{7}{2}}} + \frac{b \sqrt{\frac{1}{2}} \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \sqrt{\frac{7}{2}}} + \frac{c \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \frac{1}{2} \sqrt{\frac{1}{2}}}{8 \sqrt{\frac{7}{2}}}$$

$$= \frac{a \sqrt{\pi} \sqrt{\pi} \sqrt{\pi}}{2 \times 8 \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}} + \frac{b \sqrt{\pi} \sqrt{\pi} \sqrt{\pi}}{2 \times 8 \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}} + \frac{c \sqrt{\pi} \sqrt{\pi} \sqrt{\pi}}{2 \times 8 \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}$$

$$= \frac{\pi a}{30} + \frac{\pi b}{30} + \frac{\pi c}{30} = \boxed{\frac{\pi}{30} (a+b+c)}$$

UNIT -4 : DPP - 10

Topic : Dirichlet's Integral

Q.1 Find the volume contained in the solid region in the first octant of the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

AKTU 2014, 2022

Q.2 Find the volume of the solid surrounded by the surface  $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} + \left(\frac{z}{c}\right)^{2/3} = 1$

Q.3 Compute  $\iiint_v x^2 yz \, dx dy dz$  throughout the volume bounded by the planes  $x=0$ ,  $y=0$ ,  $z=0$

and  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

AKTU 2017

Q.4 Find the mass of solid  $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r = 1$ , the density at any point being  $\rho = kx^{l-1}y^{m-1}z^{n-1}$ ,

where  $x, y, z$  all are positive

AKTU 2016

Q.5 Find the mass of the region by ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  if the density varies as the square of the distance from the origin



## ENGINEERING MATHEMATICS - I

### UNIT -4 : Multiple Integration

#### Lecture - 11

#### Today's target

- BETA AND GAMMA FUNCTION

(Liouville's extension of Dirichlet Integral)

- PYQs
- DPP



By Gulshan sir

If  $v$  is the region bounded by  $x \geq 0, y \geq 0, z \geq 0$  and subject to the condition  $h_1 < \underbrace{(x + y + z)}_h < h_2$

then Liouville's extension of Dirichlet Integral is

$$\iiint_v f(x + y + z) x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\Gamma l \Gamma m \Gamma n}{\Gamma l+m+n} \int_{h_1}^{h_2} f(h) h^{l+m+n-1} dh$$

(1) If  $n$  is a positive integer (Natural Number)

$$\Gamma n = (n - 1)!$$

(2) If  $n$  is a positive fraction (Numerator  $>$  Denominator)

$$\Gamma n = (n - 1) \Gamma n - 1$$

**Important Results**

(i)  $\Gamma 1 = 1$

(ii)  $\Gamma \frac{1}{2} = \sqrt{\pi}$



## First Method

$$\int u v dx = u \int v dx - \int \left( \frac{d}{dx}(u) \int v dx \right)$$

## Second Method

$$\int u v dx = (u)(v_1) - (u')(v_2) + (u'')(v_3) - (u''')(v_4) + \dots$$

Gateway Classes

Q.1 Evaluate  $\iiint e^{x+y+z} dx dy dz$ , taken over positive octant such that  $x + y + z \leq 1$

Given

$$x \geq 0, y \geq 0, z \geq 0 \text{ and } \frac{0}{h_1} \leq \frac{x+y+z}{h} \leq \frac{1}{h_2}$$

$$\iiint e^{x+y+z} dx dy dz$$

$$= \iiint e^{x+y+z} x^0 y^0 z^0 dx dy dz$$

$$= \iiint e^{x+y+z} x^{1-1} y^{1-1} z^{1-1} dx dy dz$$

$$= \frac{\pi \pi \pi}{1+1+1} \int_0^1 e^h x^{1+1-1} y^{1+1-1} z^{1+1-1} dh \left\{ \begin{array}{l} \text{If } x \geq 0, y \geq 0, z \geq 0 \text{ and } h_1 \leq x+y+z \leq h_2 \text{ then} \\ \iiint f(x+y+z) x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{\pi \pi \pi}{1+m+n} \int_{h_1}^{h_2} f(h) h^{l+m+n-1} dh \end{array} \right.$$



$$\iiint e^{x+y+z} dx dy dz = \frac{1 \times 1 \times 1}{1!} \int_0^1 e^h h^2 dh$$

$$= \frac{1}{2!} \left[ (h^2)(e^h) - (2h)(e^h) + (2)(e^h) \right]_0^1$$

$$= \frac{1}{2 \times 1} \left[ e - 2e + 2e - (0 - 0 + 2) \right]$$

$$= \frac{1}{2} (e - 2)$$

Q.2 Evaluate  $\iiint \log(x + y + z) dx dy dz$ , the integral extending over all positive and zero

values of  $x, y, z$  subject to  $x + y + z < 1$

Given

$$x > 0, y > 0, z > 0 \text{ and } \frac{0}{h_1} \leq \frac{x+y+z}{h} < \frac{1}{h_2}$$

$$\iiint \log(x+y+z) dx dy dz$$

$$= \iiint \log(x+y+z) x^{1-1} y^{1-1} z^{1-1} dx dy dz$$

$$= \frac{\pi \pi \pi}{1+1+1} \int_0^1 \log h \times h^{1+1+1-1} dh \left\{ \text{If } \frac{1}{h} \right\}$$

$$= \frac{1}{13} \int_0^1 \log h \times h^2 dh \quad \text{①} \quad \text{②}$$

$$= \frac{1}{2!} \left[ \log h \int_0^1 h^2 dh - \int_0^1 \left( \frac{d}{dh} (\log h) \right) \int_0^1 h^2 dh \right]$$

$$= \frac{1}{2} \left[ \left( \log h \times \frac{h^3}{3} \right) \Big|_0^1 - \int_0^1 \frac{1}{h} \times \frac{h^3}{3} dh \right]$$

$$= \frac{1}{2} \left[ (0 - 0) - \frac{1}{3} \int_0^1 h^2 dh \right]$$



$$= \frac{1}{2} \left[ -\frac{1}{3} \left( \frac{h^3}{3} \right)'_0 \right]$$

$$= \frac{1}{2} \left[ -\frac{1}{3} \left( \frac{1}{3} \right) \right]$$

$$= -\frac{1}{18}$$

Gateway Classes

Q.3 Evaluate  $\iiint xyz \sin(x + y + z) dx dy dz$ , the integral being extended to all positive

values of the variables subject to the condition  $x + y + z \leq \frac{\pi}{2}$

AKTU 2022

Given

$$x \geq 0, y \geq 0, z \geq 0 \text{ and } \underbrace{0 \leq x+y+z}_{h_1} \leq \underbrace{\frac{\pi}{2}}_{h_2}$$

$$\iiint xyz \sin(x+y+z) dx dy dz$$

$$= \iiint \sin(x+y+z) x^{2-1} y^{2-1} z^{2-1} dx dy dz$$

$$= \frac{\Gamma 2 \Gamma 2 \Gamma 2}{\Gamma 2+2+2} \int_0^{\frac{\pi}{2}} \sinh x h^{2+2+2-1} dh \left\{ \text{I} \right\}$$

$$= \frac{1! \times 1! \times 1!}{\Gamma 6} \int_0^{\frac{\pi}{2}} \frac{\sinh h}{(2)} \times \frac{h^5}{(1)} dh$$

$$= \frac{1}{5!} \left[ (h^5)(-\cosh) - (5h^4)(-\sinh) + (20h^3)(\cosh) - (60h^2)(\sinh) + (120h)(-\cosh) - (120)(-\sinh) \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{5!} \left[ 0 + 5\left(\frac{\pi}{2}\right)^4 + 0 - 60\left(\frac{\pi}{2}\right)^2 - 0 + 120 \right]$$



$$= \frac{1}{5!} \left[ \frac{5}{16} \pi^4 - \frac{60}{4} \pi^2 + 120 \right]$$

$$= \frac{1}{5!} \left[ \frac{5}{16} \pi^4 - 15\pi^2 + 120 \right]$$

$$= \frac{1 \times \cancel{5}}{\cancel{5} \times 4 \times 3 \times 2 \times 1} \left[ \frac{\pi^4}{16} - 3\pi^2 + 24 \right]$$

$$= \frac{1}{24} \left[ \frac{\pi^4}{16} - 3\pi^2 + 24 \right]$$

**Q.4** Prove that  $\iiint \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^2}{8}$ , the integral being extended to all positive values of the variables for which the expression is real

**AKTU 2016**

$$\iiint \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^2}{8}$$

Given

$$x > 0, y > 0, z > 0 \text{ and } 0 \leq x^2 + y^2 + z^2 < 1$$

$$\iiint \frac{dx dy dz}{\sqrt{1-(x^2+y^2+z^2)}} = \frac{\pi^2}{8}$$

Put

$$x^2 = u$$

$$y^2 = v$$

$$z^2 = w$$

$$x = \sqrt{u}$$

$$y = \sqrt{v}$$

$$z = \sqrt{w}$$

$$dx = \frac{1}{2\sqrt{u}} du$$

$$dy = \frac{1}{2\sqrt{v}} dv$$

$$dz = \frac{1}{2\sqrt{w}} dw$$

Expression is real if

$$1-(x^2+y^2+z^2) > 0$$

$$x^2+y^2+z^2 < 1$$

Then

$$u > 0, v > 0, w > 0 \text{ and } 0 \leq \frac{u+v+w}{h} < \frac{1}{h_2}$$



LHS

$$\iiint \frac{dx dy dz}{\sqrt{1-(x^2+y^2+z^2)}}$$

$$= \iiint \frac{1}{\sqrt{1-(u+v+w)}} \times \frac{du}{2\sqrt{u}} \times \frac{dv}{2\sqrt{v}} \times \frac{dw}{2\sqrt{w}}$$

$$= \frac{1}{8} \iiint \frac{1}{\sqrt{1-(u+v+w)}} \times u^{-\frac{1}{2}} v^{-\frac{1}{2}} w^{-\frac{1}{2}} du dv dw$$

$$= \frac{1}{8} \iiint \frac{1}{\sqrt{1-(u+v+w)}} \times u^{\frac{1}{2}-1} v^{\frac{1}{2}-1} w^{\frac{1}{2}-1} du dv dw$$

Formulae

$$(1) \int_0^1 x^{m-1} (1-x)^{n-1} dx = \beta(m, n)$$

$$(2) \beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m + n}$$

$$= \frac{1 \times \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{8 \sqrt{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}}} \int_0^1 \frac{1}{\sqrt{1-h}} \times h^{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 1} dh$$

$$= \frac{\sqrt{\pi} \times \sqrt{\pi} \times \sqrt{\pi}}{8 \sqrt{\frac{3}{2}}} \int_0^1 h^{\frac{1}{2}} (1-h)^{-\frac{1}{2}} dh$$

$$= \frac{\pi \sqrt{\pi}}{4 \times 8 \times \frac{1}{2} \times \sqrt{\pi}} \int_0^1 h^{\frac{3}{2}-1} (1-h)^{\frac{1}{2}-1} dh$$

$$= \frac{\pi}{4} \beta\left(\frac{3}{2}, \frac{1}{2}\right)$$

$$= \frac{\pi \sqrt{\frac{3}{2}} \sqrt{\frac{1}{2}}}{4 \sqrt{\frac{3}{2} + \frac{1}{2}}}$$

$$= \frac{\pi \times \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{4 \times \sqrt{2}}$$

$$= \frac{\pi \times \sqrt{\pi} \times \sqrt{\pi}}{2 \times 4 \times 1}$$

$$= \frac{\pi^2}{8}$$

$$LHS = RHS$$

Hence proved

Gateway Classes



## UNIT -4 : DPP - 11

### Topic : Liouville's extension of Dirichlet Integral

**Q.1** Evaluate  $\iiint (x + y + z)^9 dx dy dz$ , over the region defined by  $x \geq 0, y \geq 0, z \geq 0$  and  $x + y + z \leq 1$

**Q.2** Show that  $\iiint \frac{dx dy dz}{(x+y+z+1)^3} = \frac{1}{2} \log 2 - \frac{5}{16}$  the integral being taken throughout the volume bounded by the planes  $x = 0, y = 0, z = 0$  and  $x + y + z = 1$

**Q.3** Evaluate  $\iiint \frac{dx dy dz}{\sqrt{a^2 - x^2 - y^2 - z^2}}$  the integral being extended to all positive values of the variables for which the expression is real.

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All Subjects

Link in Description

**Thank You**